TM Forum Specification

Service Problem Management User Guide

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Introduction

This Service Problem Management API enables service providers to manage service related issues within their service area. Service problems may originate from two sources:

• Service Providers, who may declare issues based on their own monitoring or customer feedback.

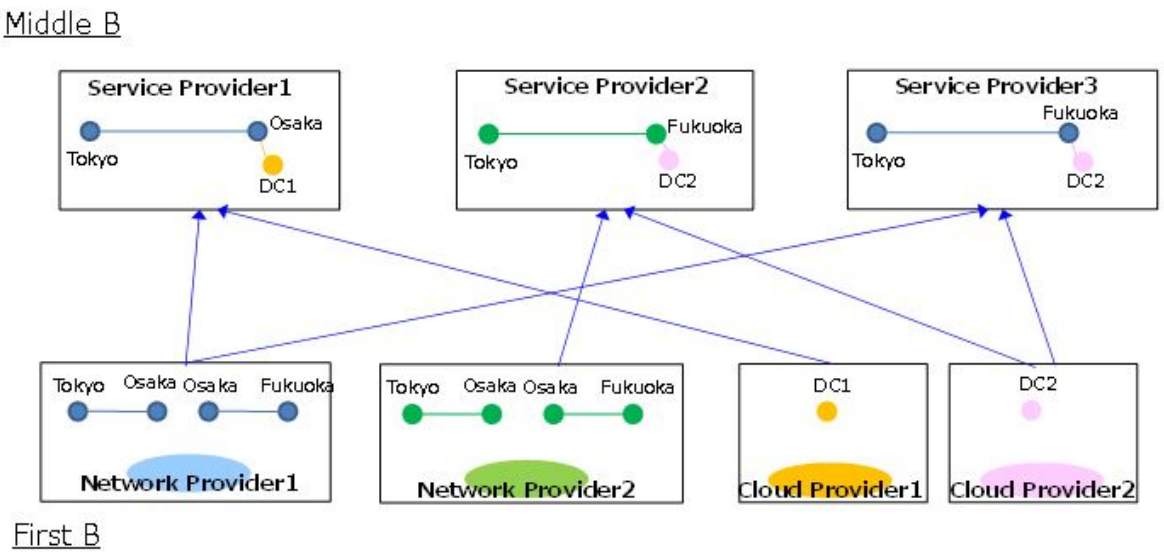
• Infrastructure Providers (e.g., cloud or network providers), who notify event information to the service providers. The event information includes alarm information, performance anomaly information, trouble ticket information, SLA violations, maintenance activities and prediction information.

Service providers can refer the service problems and the event information from infrastructure providers and receive notifications when new service problems arise or when the status of existing problems changes. This functionality allows service providers to detect and respond to service issues quickly and accurately, improving operational efficiency and customer satisfaction.

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Sample Use Cases

We assume following situation: There are Network Provider 1 (NP1) and 2 (NP2), which provide network infrastructure, and Cloud Provider 1, 2, which provides cloud infrastructure, as First Bs. Using this infrastructure, Service Provider 1 (SP1), 2 (SP2) and 3 (SP3) are providing their services to their end-users as Middle Bs.

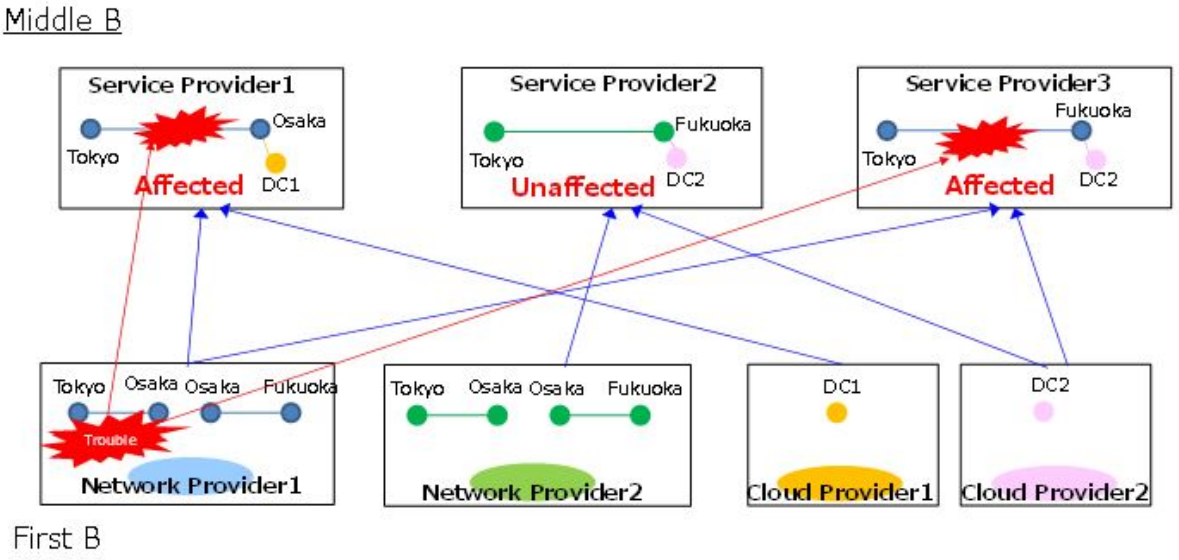


Use Case 1

When trouble happen in any resources of NW/Cloud Providers, Service Providers can know if their services are affected or not. The specific use case is following:

1. SPM collects configuration information of services provided by service providers in advance using Product Inventory Management API, etc.
2. Each of Middle Bs – Service Provider 1 (SP1), Service Provider 2 (SP2), Service Provider 3 (SP3) – registers the notification destination to SPM SPI.
3. When a fault occurs, SPM receives a trouble ticket from the Network Provider 1 (NP1).
4. SPM creates a Service Problem based on the Trouble Ticket.
5. SPM notifies the Service Problem creation notification to Middle B (SP1, SP3) to notify expected service impact, based on the configuration information collected in advance.
6. When SPM receives a notification that the trouble ticket has changed to "In Progress" state, update the status of the relevant Service Problem. Notify the Service Problem state change notification to Middle B (SP1, SP3).

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Use Case 2

To analyze the past problems, Middle B collects the problem information in the past one year

Use Case 3

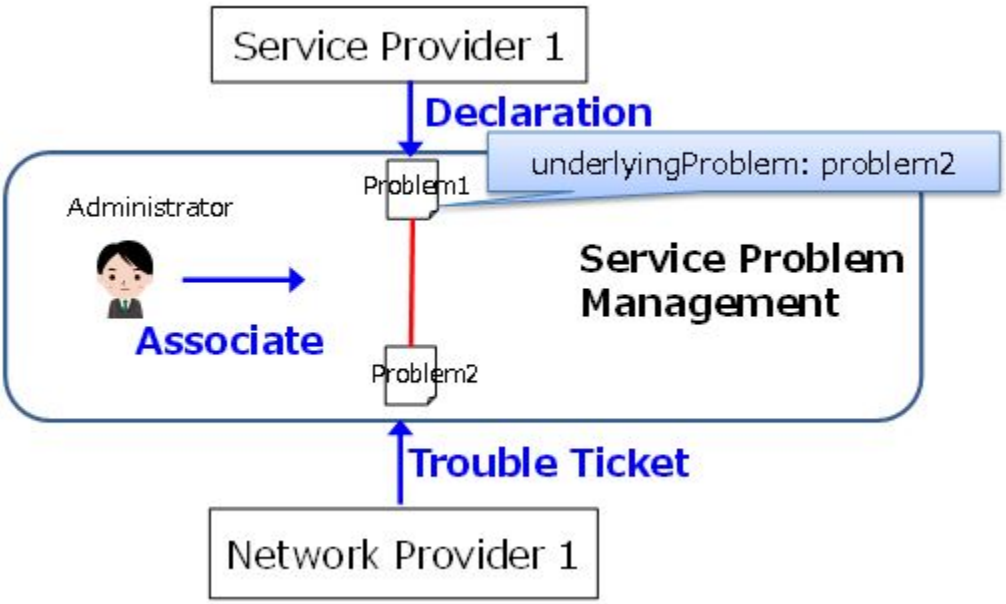
Service providers can declare a new service problem based on trouble declarations from their end-users. In addition, the SPM administrator can associate the service problem, based on the Middle B declaration, with another problem based on a First B event such as a Trouble Ticket. The specific use case is following:

1. Based on the report from the user that there is a problem in the Internet access, Middle B (SP1) gets the current service problem.
2. After SPM collects the current Service Problems, returns that there are no problems related to the service of the Middle B (SP1).
3. In order to request the analysis of this event, Middle B declares a new service problem.
4. Since the SPM administrator found that necessary detailed information was insufficient, SPM administrator requests additional information about the behavior of the Middle B side.
5. SP1 collects the specified additional information and registers it.
6. SPM administrator checks the additional information and accepts the Service Problem (Problem 1).
7. First B(NP1) registers a detected problem event to the trouble ticket and notice a new

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trouble ticket generation to SPM. The SPM creates a Service Problem (Problem 2) based on the trouble ticket.

8. Since the two problems affects the same location, SPM administrator determines that the declared problem (Problem 1) and the new problem based on the new trouble ticket (Problem 2) is associated. SPM administrator associates Problem 1 with 2. In this case, Problem 1 will have Problem2 as the association “underlyingProblem”. Note that Problem 1 can have an “parentProblem” as another association if he would like to group those problems.
9. Since the Problem 1 was changed to add a “underlyingProblem”, Service Problem Change Notification is sent to SP1.

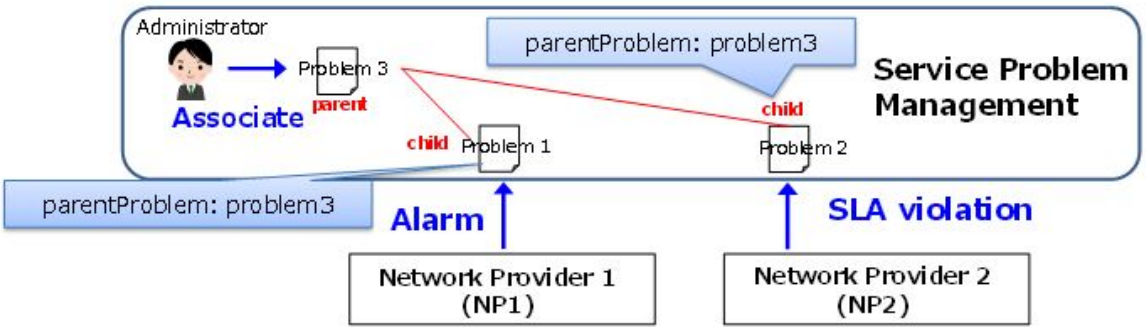


Use Case 4

The SPM administrator can associate and group multiple service problems so that service providers can easily recognize what the real problem is. The specific use case is following:

1. SPM receives an alarm from NP1 and creates a Service Problem based on it (Problem 1).
2. SPM receives an SLA violation from NP2 and creates a Service Problem based on it (Problem 2).
3. By analyzing problems, SPM administrator determines that Problem 1 and 2 are the same problem. SPM administrator creates a new Service Problem (Problem 3) in order to group and associate Problem 1, 2 and 3. In this case, Problem 3 is a parent and Problem 1 and 2 are children.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a TypeAServiceProblem and TypeBServiceProblem inheriting properties from the abstract ServiceProblem entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v5.0 Part 2 document.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of ServiceProblem instances some may be instances of TypeAServiceProblem where other could be instances of TypeBServiceProblem. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an RelatedParty object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as ServiceRef or TroubleTicketRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

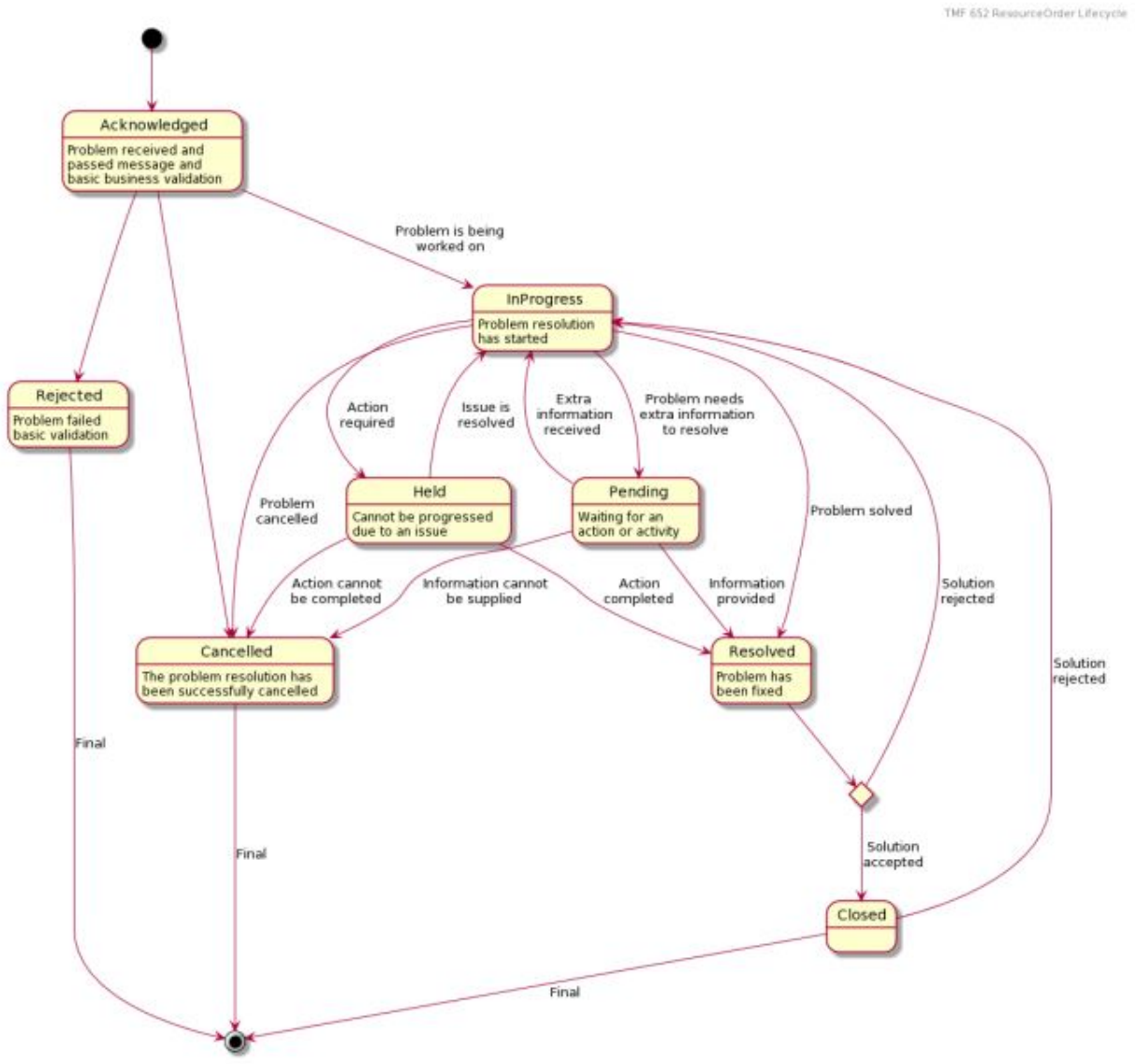
The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

ServiceProblem status lifecycle

The state diagram below gives an overview of the allowed status changes



The following table provides the service problem status explanation:

Status	Definition
Acknowledged	The Service Problem is acknowledged by Service problem handler.

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Status	Definition
Rejected	The Service Problem was Rejected because of • Invalid or inadequate information is provided • Failes to meet the business rules with respect to the service which consumer is raising a service problem against • Is otherwise defective
InProgress	The service problem was validated by the service problem handler and is being processed.
Held	Service problem handler is confirming further details internally before completing a service problem.
Pending	Service problem handler is awaiting further confirmation on details of a Fault from originator before it can progress the Fault.
Resolved	The fault indicated in the service problem was corrected by the service problem handler and acknowledgement is awaited from its originator.
Closed	The service Problem's originator has acknowledged the 'Resolved' state of the service problem, or the timeframe for acknowledgement has passed without response from service problem originator.
Cancelled	The service problem was cancelled due to the following reasons: • The service problem originator cancelled • The service problem handler cancelled as the problem was not related to the handler • The service provider handler deemed that the problem cannot be resolved (non-fixable problem)

ProblemGroup and ProblemUngroup status lifecycle

ProblemGroup and ProblemUngroup operations follow the Task lifecycle model, with states including acknowledged, rejected, inProgress, cancelled, done and terminatedWithError

Managed Entity and Task Resource Models

ServiceProblem resource

A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected

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performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations.

Resource model

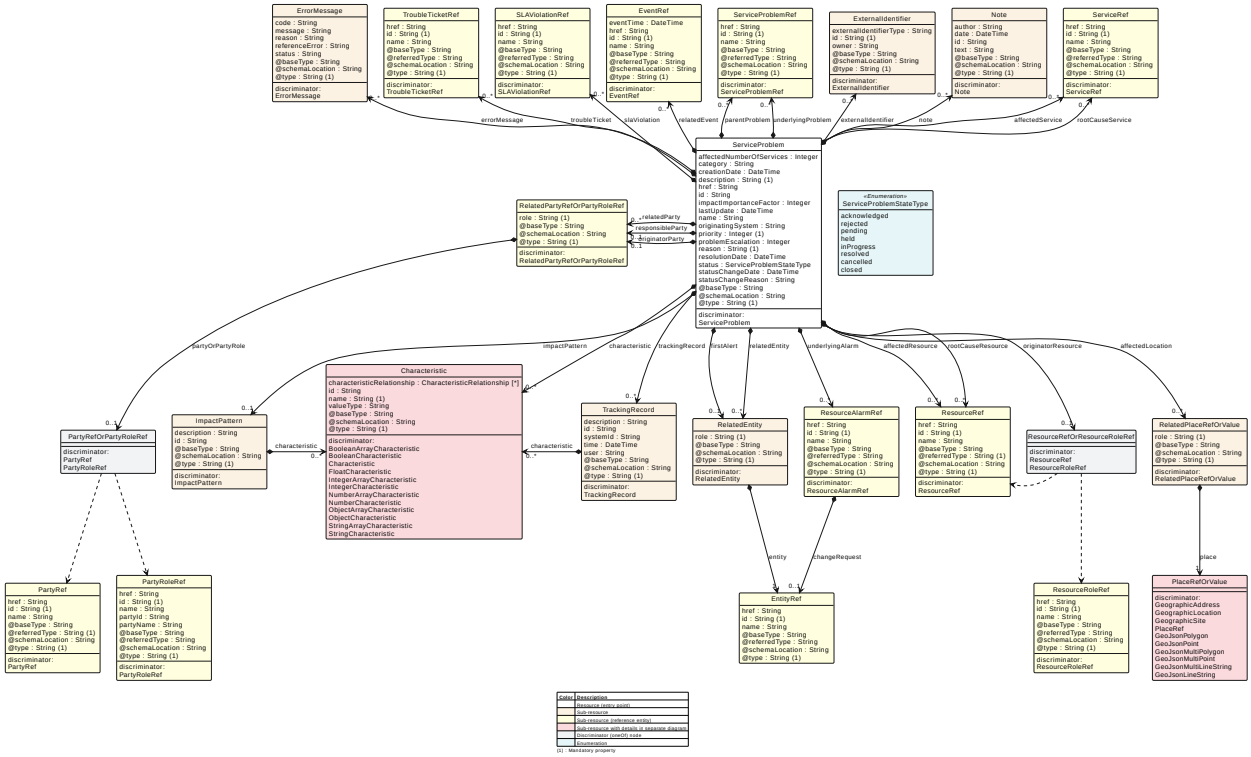


Figure 1 - ServiceProblem

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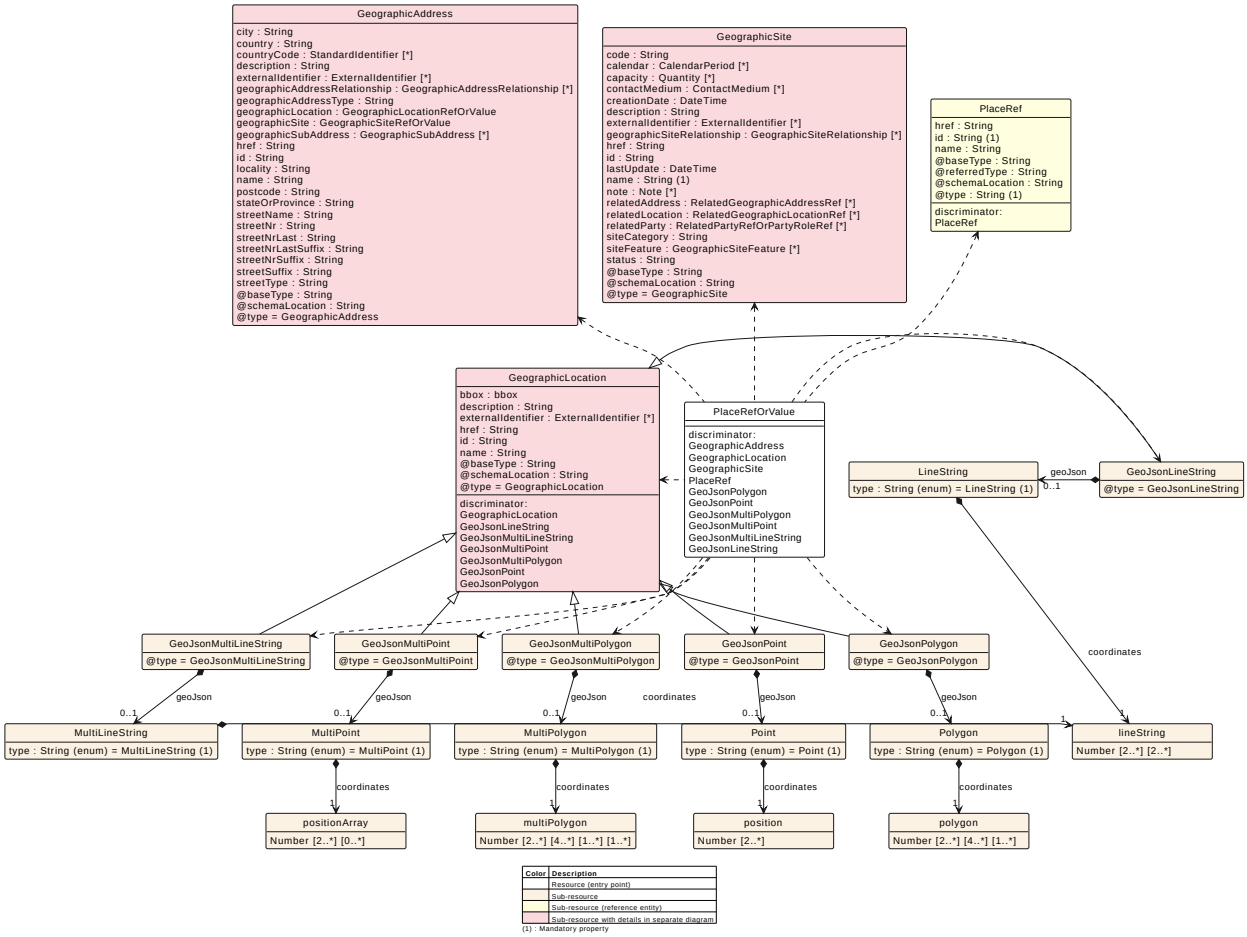


Figure 2 - PlaceRefOrValue

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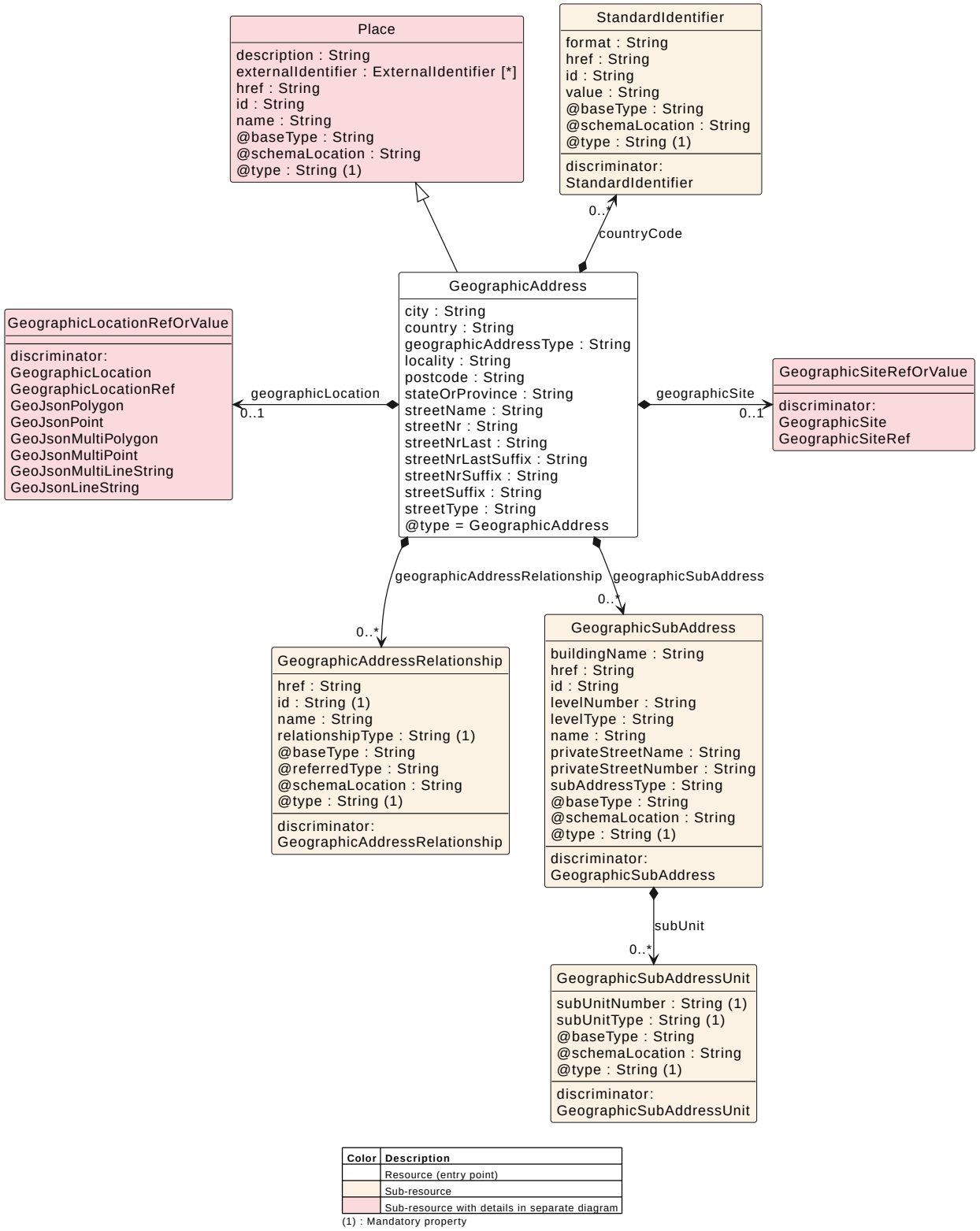


Figure 3 - GeographicAddress

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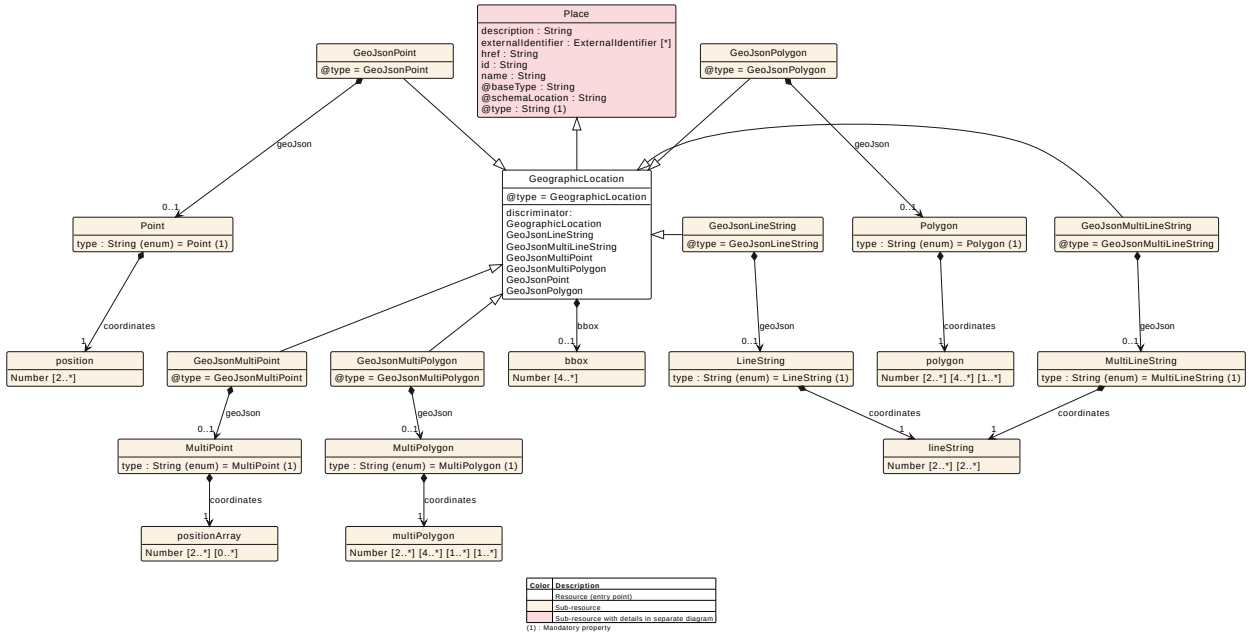


Figure 4 - GeographicLocation

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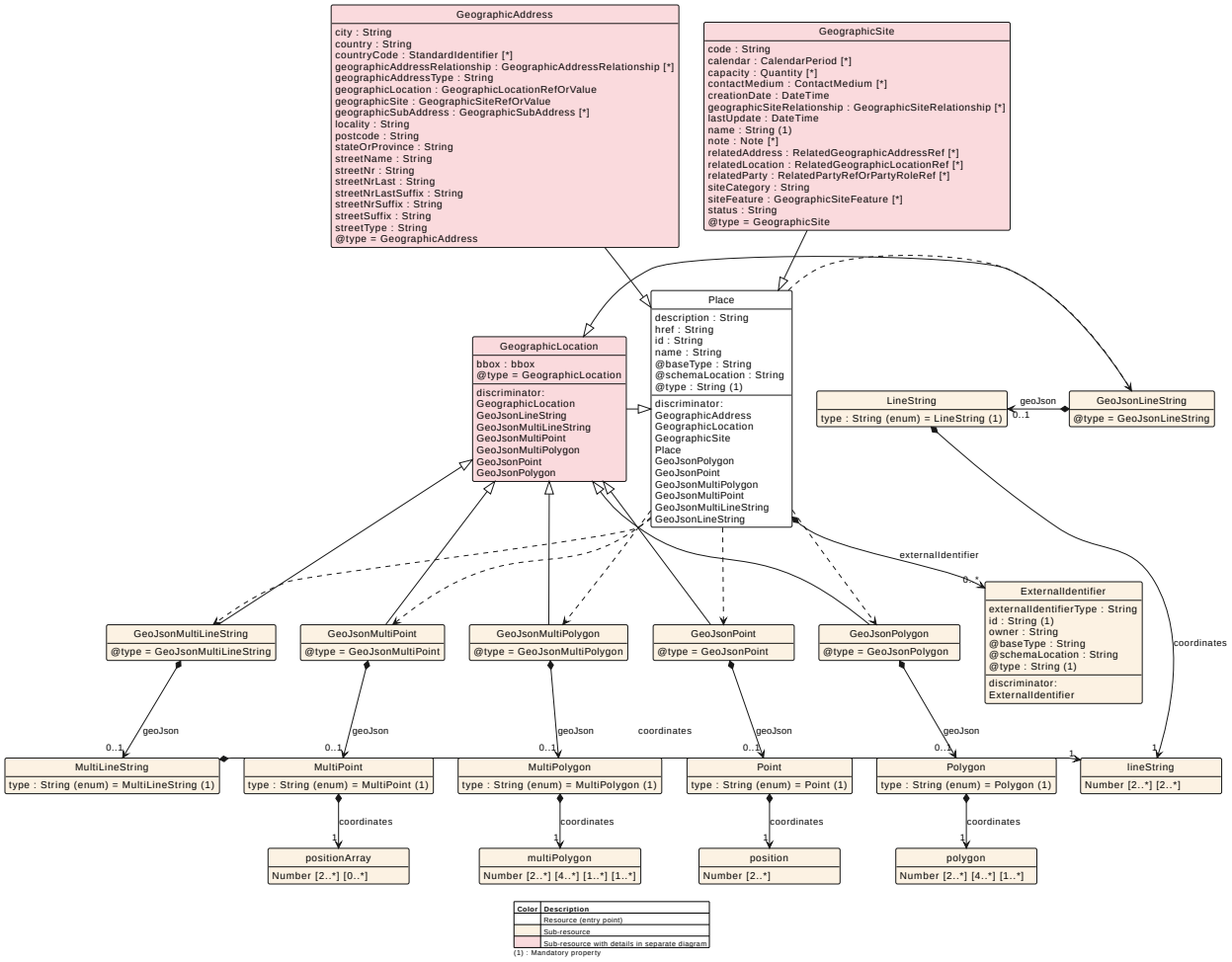


Figure 5 - Place

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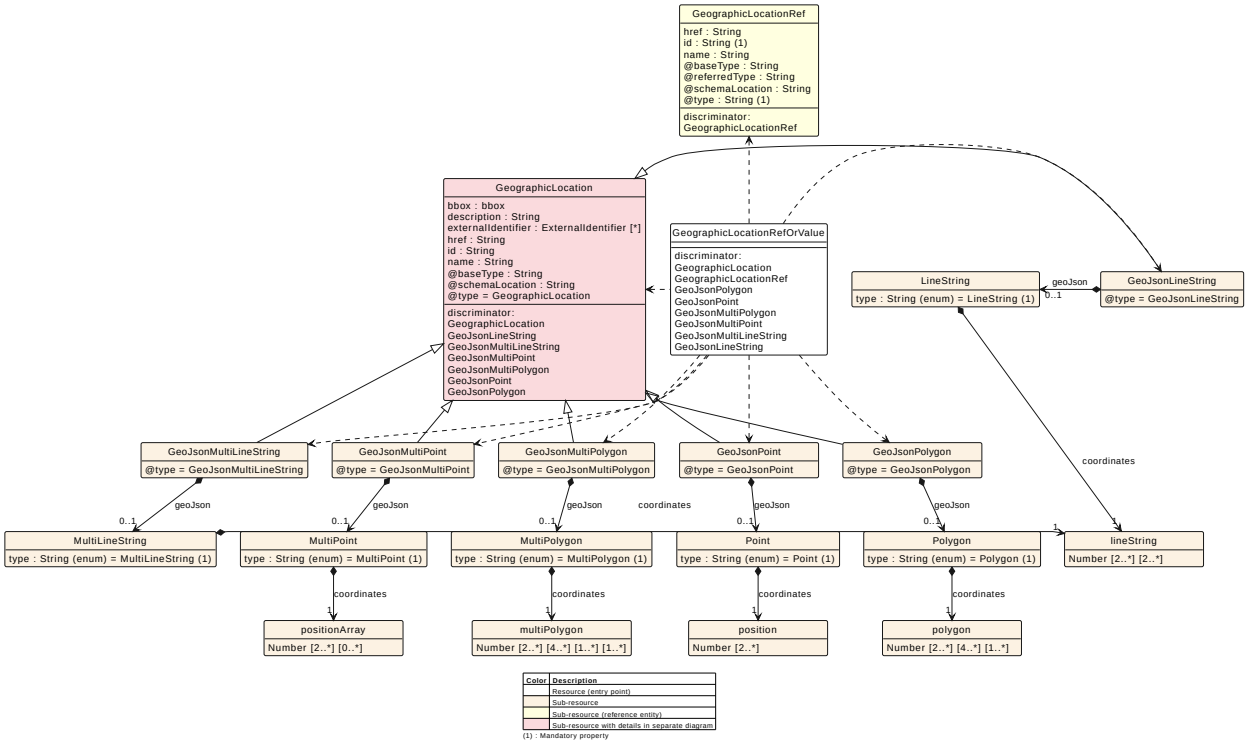


Figure 6 - GeographicLocationRefOrValue

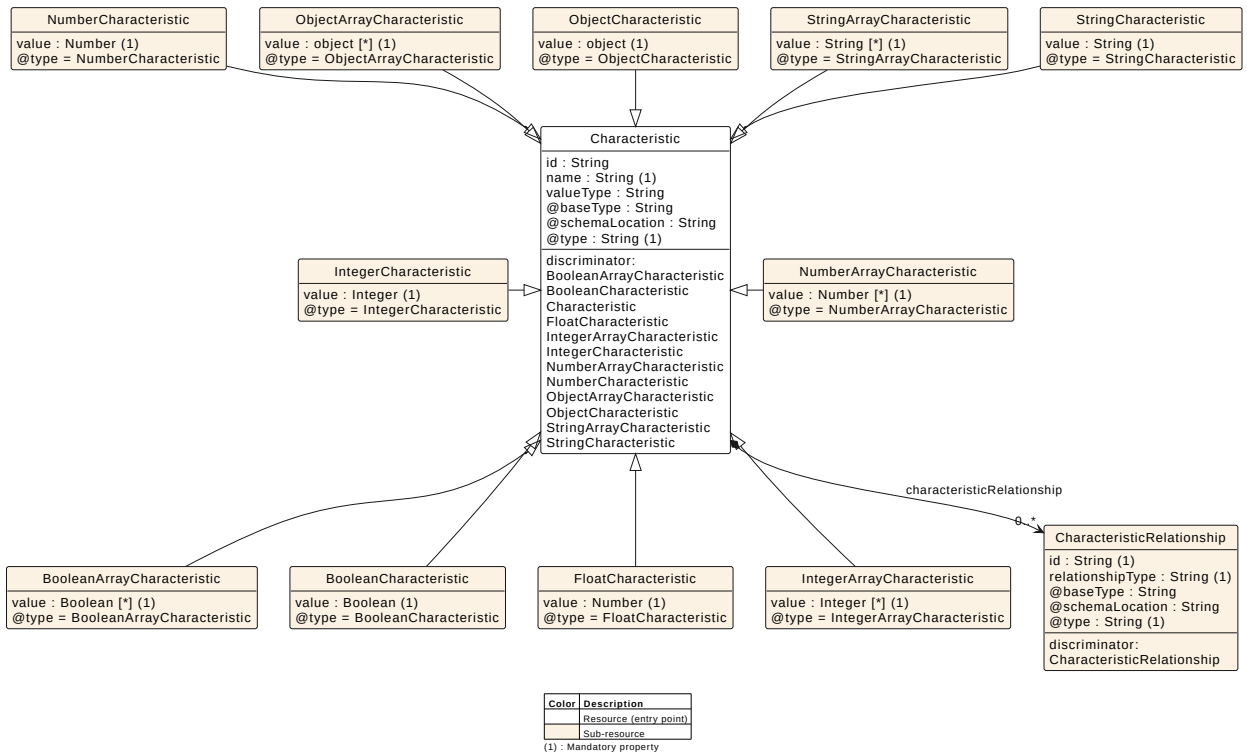


Figure 7 - Characteristic

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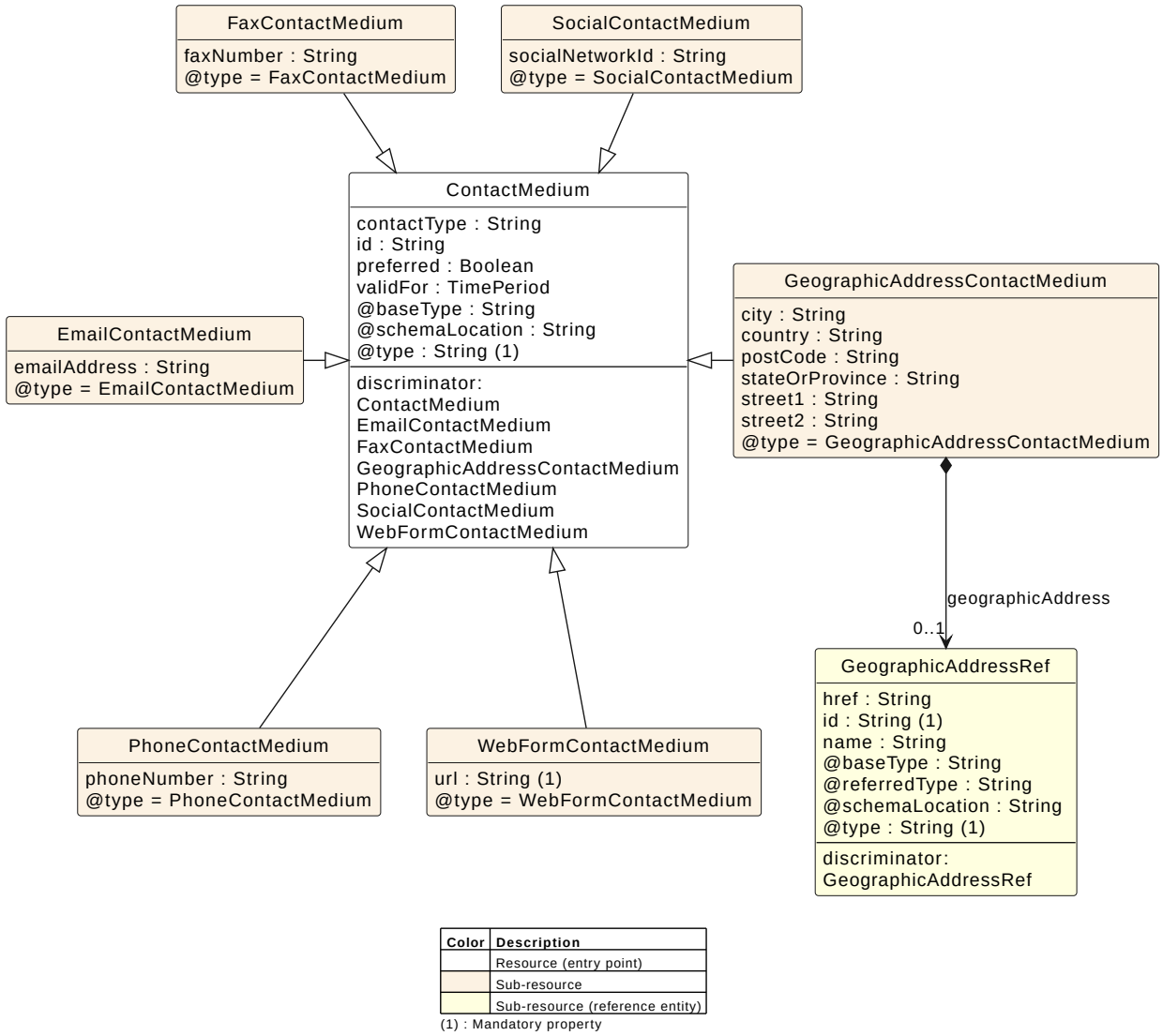


Figure 8 - ContactMedium

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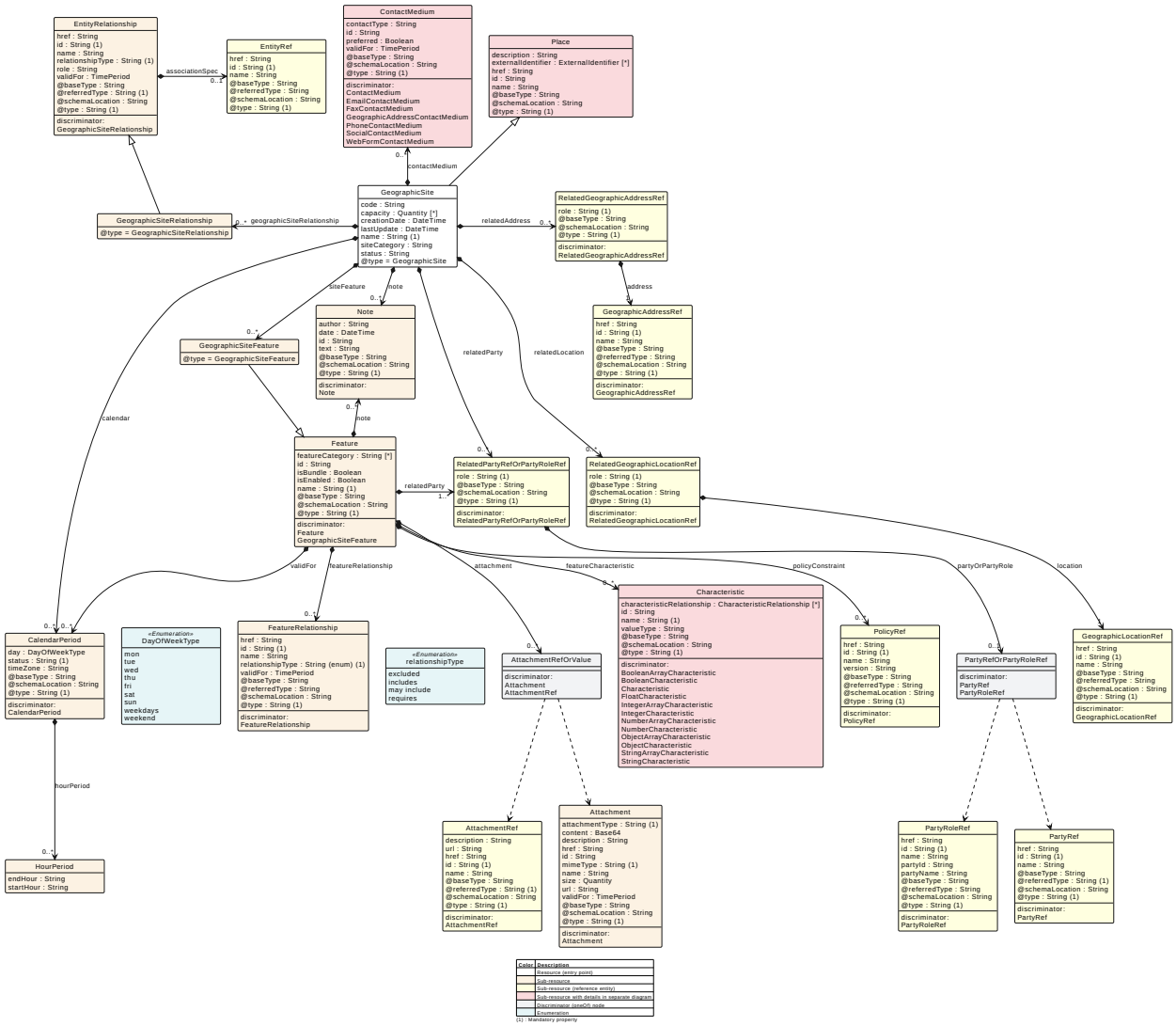


Figure 9 – GeographicSite

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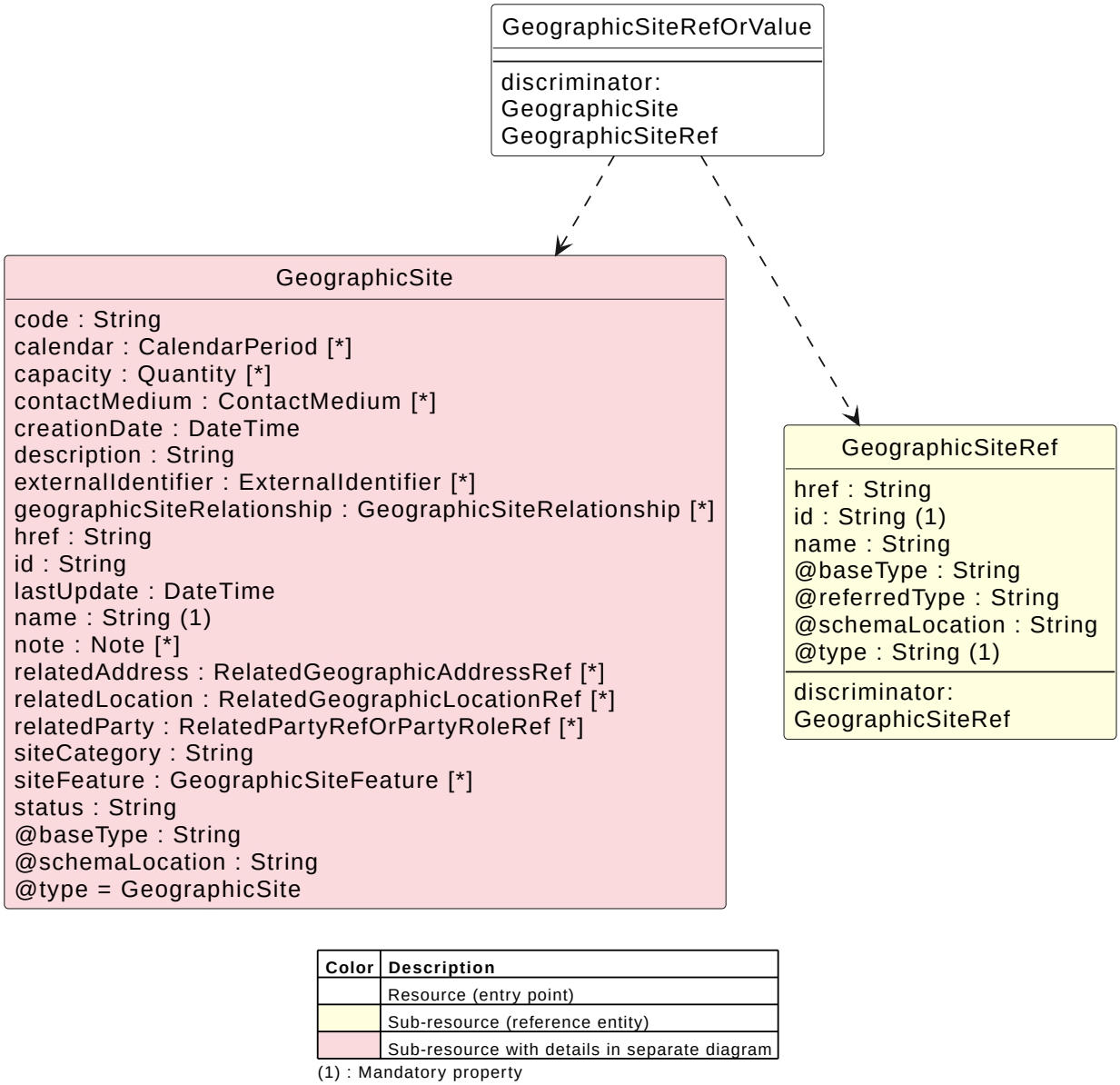


Figure 10 - GeographicSiteRefOrValue

Field descriptions

ServiceProblem fields

affectedLocation	A RelatedPlaceRefOrValue [0..*]. A list of the locations affected by the problem. At least one of affected Resource, affectedService or affectedLocation should be present.
affectedNumberOfServices	An Integer. Number of affected services.

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affectedResource	A ResourceRef [0..*]. A list of the resources affected by the problem. At least one of affectedResource, affected Service or affectedLocation should be present.
affectedService	A ServiceRef [0..*]. List of affected services. At least one of affectedResource, affectedService or affectedLocation should be present.
category	A String. Classifier for the problem. Settable. For example, this is used for distinguish the category of problem originator in [role].[category] format. Example: service Provider.declarer, supplier.originated, system.originated.
characteristic	A Characteristic [0..*]. A list of additional properties/data items that describe or characterize this service problem. Please note that this list is currently unconstrained but in a future version this property might be constrained by a specification or deprecated altogether. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
creationDate	A DateTime. Time the problem was created.
description	A String [1]. Free form text describing the Service Problem.
errorMessage	An ErrorMessage [0..*]. A list of the error(s) associated to the Service Problem. This object is used for any asynchronous errors and error information to be shared with consumer.
externalIdentifier	An ExternalIdentifier [0..*]. A list of external identifiers assoicated with this problem.
firstAlert	A RelatedEntity. A reference to an entity, where the type of the entity is not known in advance.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
impactImportanceFactor	An Integer. Impact Importance is characterized by an Impact Importance Factor: overall importance of the impact of all the affected services, e.g. 0 (zero impact) to 100 (worst impact). The Impact Importance is a calculated field which is set by the OSS determining the impact.
impactPattern	An ImpactPattern. ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.
lastUpdate	A DateTime. Time the problem was last changed.
name	A String. Name of the Service Problem.
note	A Note [0..*]. A list of comments or notes made on the problem.
originatingSystem	A String. Indicates in which system the problem was generated. Example: CRM Systems, Assurance System, etc.
originatorParty	A RelatedPartyRefOrPartyRoleRef. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
originatorResource	A ResourceRefOrResourceRoleRef. Resource reference. A reference to a Resource or to the role that a resource plays in a specific context.
parentProblem	A ServiceProblemRef [0..*]. The parent problem to which this problem is attached.
priority	An Integer [1]. An indication varying from 1 (highest) to 10 (lowest) of how important it is for the service provider to correct the Service Problem.
problemEscalation	An Integer. Indicates if this service problem has been escalated or not. Possible values are 0 to 10. A value of zero means no escalation. The meanings of values 1-10 are to be determined by the user of the interface, but they show increasing levels of escalation.

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reason	A String [1]. Reason, if known, for the service problem that will assist the Service Provider in understanding the underlying cause or contributing factors, enabling more accurate diagnosis and resolution. This can be set to Unknown, otherwise.
relatedEntity	A RelatedEntity [0..*]. List of entities associated with this problem. Example: Reference to TroubleTicket, Alarm etc.
relatedEvent	An EventRef [0..*]. List of events associated to this problem.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. List of parties or party roles playing a role within the service problem.
resolutionDate	A DateTime. Time the problem was resolved.
responsibleParty	A RelatedPartyRefOrPartyRoleRef. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
rootCauseResource	A ResourceRef [0..*]. Resource(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
rootCauseService	A ServiceRef [0..*]. Service(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
slaViolation	A SLAViolationRef [0..*]. A List of SLA violations associated with this problem.
status	A ServiceProblemStateType. The life cycle status of the service problem, such as acknowledged, rejected, inProgress, resolved etc... ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * resolved * cancelled * closed
statusChangeDate	A DateTime. Time the problem was last status changed.
statusChangeReason	A String. The reason of state change.

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trackingRecord	A TrackingRecord [0..*]. List of tracking records that allow the tracking of modifications on the problem.The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
troubleTicket	A TroubleTicketRef [0..*]. A list of trouble tickets associated with this problem.
underlyingAlarm	A ResourceAlarmRef [0..*]. A list of alarms underlying this problem.
underlyingProblem	A ServiceProblemRef [0..*]. A list of underlying problems. Relevant only if this problem is derived from other problems.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

Use `role` to distinguish the context of the Place being referenced (such as a `delivery-entrance`). In the context of a Ref, use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). In the context of a Value, use `@type` to specify the Place sub-type.

place	A PlaceRefOrValue [1]. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String [1]. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanProperty

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Address.

city	A String. City that the address is in.
country	A String. Country that the address is in.
countryCode	A StandardIdentifier [0..*]. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geographicAddressRelationship	A GeographicAddressRelationship [0..*]. A list of geographic addresses relationships. Represents a relationship between geographic addresses.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress [0..*]. Collection of sub addresses.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
name	A String. A user-friendly name for the Place.
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource. It is anticipated that multiple standards can provide definitions for a single entity, e.g., a country identifier can be specified in various standards (e.g., "ISO 3166-1 Alpha 2", "ISO 3166-1 Alpha 3", "ISO 3166-1 Numeric").

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format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String [1]. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

The GeographicAddressRelationship schema represents a relationship between geographic

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addresses. It defines the structure for storing information about how two geographic addresses are related to each other within a system.

href	A String. The URI of the referred entity.
id	A String [1]. Unique identifier of the related geographic address. This identifier serves as a reference to uniquely identify the associated geographic address within the system.
name	A String. Name of the geographic address. This attribute provides a human-readable name or label for the related geographic address, which can help users identify and differentiate between different addresses.
relationshipType	A String [1]. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

A GeographicLocation is a pure-virtual super-class to the GeoJSON-aligned geometries of Point (addresses and locations), MultiPoint, LineString (streets, highways and boundaries), MultiLineString and Polygon (countries, provinces, tracts of land). Use the @type attribute to specify which of these is being specified by the geometry attribute.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.

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externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	GeographicLocation can be instantiated as * GeographicAddress * GeographicSite * Place * GeoJsonPolygon * GeoJsonPoint * GeoJsonMultiPolygon * GeoJsonMultiPoint * GeoJsonMultiLineString * GeoJsonLineString
value	This property is present in subclasses

GeographicLocationRef sub-resource fields

Reference to a Geographic Location.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeoJsonPolygon sub-resource fields

A GeoJsonPolygon is an array of linear rings.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A Polygon. GeoJSON: An array of linear rings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Polygon sub-resource fields

GeoJSON: An array of linear rings.

coordinates	A polygon [1]. GeoJSON: An array of linear rings.
type	A String [1]. ENUMERATED with values: * Polygon

GeoJsonPoint sub-resource fields

A GeoJsonPoint is a single point.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.

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description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A Point. GeoJSON: A single position.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Point sub-resource fields

GeoJSON: A single position.

coordinates	A position [1]. GeoJSON: A single position.
type	A String [1]. ENUMERATED with values: * Point

GeoJsonMultiPolygon sub-resource fields

A GeoJson MultiPolygon is an array of polygons.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiPolygon. GeoJSON: A collection of Polygons.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MultiPolygon sub-resource fields

GeoJSON: A collection of Polygons.

coordinates	A multiPolygon [1]. GeoJSON: An array of polygons.
type	A String [1]. ENUMERATED with values: * MultiPolygon

GeoJsonMultiPoint sub-resource fields

GeoJson MultiPoint is a one or more GeoJSON points.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiPoint. GeoJSON: A collection of Points.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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MultiPoint sub-resource fields

GeoJSON: A collection of Points.

coordinates	A positionArray [1]. GeoJSON: An array of positions.
type	A String [1]. ENUMERATED with values: * MultiPoint

GeoJsonMultiLineString sub-resource fields

GeoJson MultiLineString is a one or more GeoJSON LineStrings.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiLineString. GeoJSON: A collection of distinct LineStrings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MultiLineString sub-resource fields

GeoJSON: A collection of distinct LineStrings.

coordinates	A lineString [1]. GeoJSON: An array of two or more positions.
type	A String [1]. ENUMERATED with values: * MultiLineString

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GeoJsonLineString sub-resource fields

GeoJson LineString is a collection of points forming a connected line.

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A LineString. GeoJSON: A collection of Points forming a connected line.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

LineString sub-resource fields

GeoJSON: A collection of Points forming a connected line.

coordinates	A lineString [1]. GeoJSON: An array of two or more positions.
type	A String [1]. ENUMERATED with values: * LineString

GeographicSite sub-resource fields

A GeographicSite is a managed place, such as a retail store, warehouse or data centre. It will have related responsible parties, a calendar of access times and be described by addresses and/or locations.

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calendar	A CalendarPeriod [0..*]. Periods of (for example) access, opening, alarm, unavailability of this site.
capacity	A Quantity [0..*]. Capacities for the site indicating maximum operational thresholds (for example: people, floorspace, racks).
~~code~~	DEPRECATED A String. Deprecated: Use id, externalIdentifier and/or site Category or name.
contactMedium	A ContactMedium [0..*]. Contact information for people with responsibility for this site. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geographicSiteRelationship	A GeographicSiteRelationship [0..*]. This site might have relationships to other sites.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time when the GeographicSite was last updated.
name	A String [1]. A user-friendly name for the Place.
note	A Note [0..*]. Unstructured notes made about this site.
relatedAddress	A RelatedGeographicAddressRef [0..*]. Collection of related address references (such as delivery-address, office-address).

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relatedLocation	A RelatedGeographicLocationRef [0..*]. Collection of related location references (such as retail-complex, service-access).
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related roles and parties references (such as site manager, site owner, site operator).
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature [0..*]. Collection of site features.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

day	A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values: * mon * tue * wed * thu * fri * sat * sun * weekdays * weekend
hourPeriod	A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format.

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status	A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

Hour interval.

endHour	A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

ContactMedium sub-resource fields

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContact Medium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
value	This property is present in subclasses

EmailContactMedium sub-resource fields

Describes an email that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

Describes a fax number that could be used to contact a party (an individual or an organization).

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contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressContactMedium sub-resource fields

Describes a geographical address that could be used to contact a party (an individual or an organization).

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.
id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRef sub-resource fields

Reference to a Geographic Address.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

Describes a phone number that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

SocialContactMedium sub-resource fields

Describes a social media identifier that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

WebFormContactMedium sub-resource fields

Describes an URL of a web form that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.

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url	A String [1]. Url of the web form.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

Details of a geographic site relationship.

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String [1].
name	A String.
relationshipType	A String [1]. Type of relationship between GeographicSites, such as adjacentTo, connectedBy.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1].
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1].

EntityRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

Extra information about a given entity.

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedGeographicAddressRef sub-resource fields

A Postal Address (by reference) with its associated role.

address	A GeographicAddressRef [1]. Reference to a Geographic Address.
role	A String [1]. The context in which this Address is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedGeographicLocationRef sub-resource fields

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A Geographic Location (by reference) with its associated role.

location	A GeographicLocationRef [1]. Reference to a Geographic Location.
role	A String [1]. The context in which this location is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

partyOrPartyRole	A PartyRefOrPartyRoleRef. A reference to a Party or a PartyRole.
role	A String [1]. The role the referred Party or PartyRole plays in the specific context of the entity it is linked to, such as 'initiator', 'customer', 'salesAgent', 'user'; NOTE: If used to refer to a PartyRole, the PartyRole describes the role the respective engagedParty plays in a more general context, like e.g. 'Customer', 'Manager'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

GeographicSiteFeature sub-resource fields

Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.

attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.

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featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

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attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

~~description~~	DEPRECATED A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6.

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href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
~~url~~	DEPRECATED A String. Link to the attachment media/content. DEPRECATED: will be removed in v6.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

id	A String [1]. Unique identifier of the characteristic.
relationshipType	A String [1]. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1..*]. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Float [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1..*]. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

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characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1..*]. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

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characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType [1..*]. A characteristic which value is an array of Object(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType_2 [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

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characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1..*]. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

Configuration feature.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. This is the name of the target feature.

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relationshipType	A String [1]. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

Reference to managed Policy object.

href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

Configuration feature.

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attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.
featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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	Feature can be instantiated as * GeographicSiteFeature
value	This property is present in subclasses

GeographicSiteRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

Representation of a GeographicSubAddress It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Link to the subAddress.
id	A String. Unique Identifier of the subAddress.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.

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privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit [0..*]. Representation of a SubUnit. It is used for describing subunit within a sub Address e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.

subUnitNumber	A String [1]. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String [1]. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

A reference to a physical Place.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The specific sub-type of Place being referred to (such as a `GeographicAddress`).
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceRef sub-resource fields

Service reference, for when Service is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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ErrorMessage sub-resource fields

Represents an Error.

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedEntity sub-resource fields

A reference to an entity, where the type of the entity is not known in advance.

entity	An EntityRef [1]. Entity reference schema to be use for all entityRef class.
role	A String [1].
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ImpactPattern sub-resource fields

ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.

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characteristic	A Characteristic [0..*]. A generic list of any type of elements. Used for extensions or loose element encapsulation from other namespaces. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
description	A String. Basic description of the impact pattern.
id	A String. Locally unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRoleRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemRef sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

EventRef sub-resource fields

Events linked with service problem.

eventTime	A DateTime. Time the event occurred.
href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

SLAViolationRef sub-resource fields

Entity reference schema to be use for all entityRef class.

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href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

TrackingRecord sub-resource fields

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

characteristic	A Characteristic [0..*]. A generic list of any type of elements. Used for vendor Extensions or loose element encapsulation from other namespaces. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
description	A String. Describes the action being done, such as: ack, clear.
id	A String. Identifier of the TrackingRecord.
systemId	A String. Describes the system Id from which the action was done.

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time	A DateTime. Describes the time at which the action was done.
user	A String. Describes the user doing the action.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

TroubleTicketRef sub-resource fields

Reference to an TroubleTicket object.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceAlarmRef sub-resource fields

A set of alarm ids identifying the alarms that are underlying this problem.

changeRequest	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ServiceProblem' resource object.


```
{
  "id": "problemxxxx0000",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": 0,
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": 0,
  "statusChangeDate": "2024-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2024-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": 2,
  "firstAlert": {
    "@type": "RelatedEntity",
    "role": "trigger",
    "entity": {
      "@type": "EntityRef",
      "@referredType": "TroubleTicket",
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/troubleTicketManagement/v5/troubleTicket/NP1_TT_000000"
    }
  },
  "responsibleParty": {
    "role": "Customer",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "456",
      "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  },
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
```


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```
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "789",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/789",
        "name": "NP1",
        "@referredType": "Organization"
      }
    },
    {
      "role": "Partner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "123",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "name": "SP1",
        "@referredType": "Organization"
      }
    },
    {
      "role": "Partner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "321",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/321",
        "name": "SP2",
        "@referredType": "Organization"
      }
    }
  ],
  "affectedService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
      "@type": "ServiceRef",
      "@referredType": "Service"
    },
    {
      "id": "NP1_Tokyo_xxxx",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
```


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```
  "affectedResource": [
    {
      "id": "NP1_RES_0001",
      "href": "https://mycsp.com:8080/tmf-
api/resourceInventory/v5/resource/NP1_RES_0001",
      "@type": "ResourceRef",
      "@referredType": "VirtualResource"
    }
  ],
  "affectedLocation": [
    {
      "role": "VPN Endpoint",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556"
      }
    },
    {
      "role": "VPN Endpoint",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365"
      }
    }
  ],
  "troubleTicket": [
    {
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/troubleTicketManagement/v5/TroubleTicketRef/NP1_TT_000000",
      "@type": "TroubleTicketRef",
      "@referredType": "TroubleTicket"
    }
  ],
  "underlyingAlarm": [
    {
      "id": "NP1_A_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/alarmManagement/v5/resourceAlarm/NP1_A_000000",
      "@referredType": "ResourceAlarm",
      "@type": "ResourceAlarmRef"
    }
  ],
  "slaViolation": [
    {
      "id": "NP1_SLA_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/SLAManagement/v5/SLAViolationRef/NP1_SLA_000000",
      "@type": "SLAViolationRef",
      "@referredType": "SLAViolation"
```


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```
    }
  ],
  "relatedEvent": [
    {
      "id": "prediction_0001",
      "href": "https://mycsp.com:8080/tmf-
api/eventManagement/v5/event/prediction_0001",
      "eventTime": "2024-10-28T17:00:00Z",
      "@type": "EventRef",
      "@referredType": "Event"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-
api/resourceInventory/v5/resource/NP1_Resource_1",
      "@type": "ResourceRef",
      "@referredType": "VNF"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "trackingRecord": [
    {
      "description": "System cleared the problem",
      "systemId": "SNOW",
      "time": "2024-10-29T12:00:00Z",
      "user": "NPUSER1",
      "id": "TR_0001",
      "@type": "TrackingRecord"
    }
  ],
  "note": [
    {
```


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```
      "id": "2241",
      "author": "SPM IT Team",
      "date": "2024-10-29T11:30:00Z",
      "text": "receive trouble ticket from NP1, and create this Service Problem",
      "@type": "Note"
    }
  ],
  "impactPattern": {
    "description": "Many services are at risk in this problem",
    "id": "TR_0002",
    "@type": "ImpactPattern",
    "characteristic": [
      {
        "id": "CH_003",
        "name": "ServiceDisruption",
        "value": true,
        "valueType": "boolean",
        "@type": "BooleanCharacteristic"
      }
    ]
  },
  "@type": "ServiceProblem"
}
```


ServiceProblemEventRecord resource

ServiceProblemEventRecord is a record of an event (related to a service problem) received from another system. It is a system-generated record that captures events related to a ServiceProblem, such as status changes, updates, acknowledgements, or resolution activities. It is not created directly by API consumers and serves as an audit and traceability mechanism for problem management activities.

Resource model

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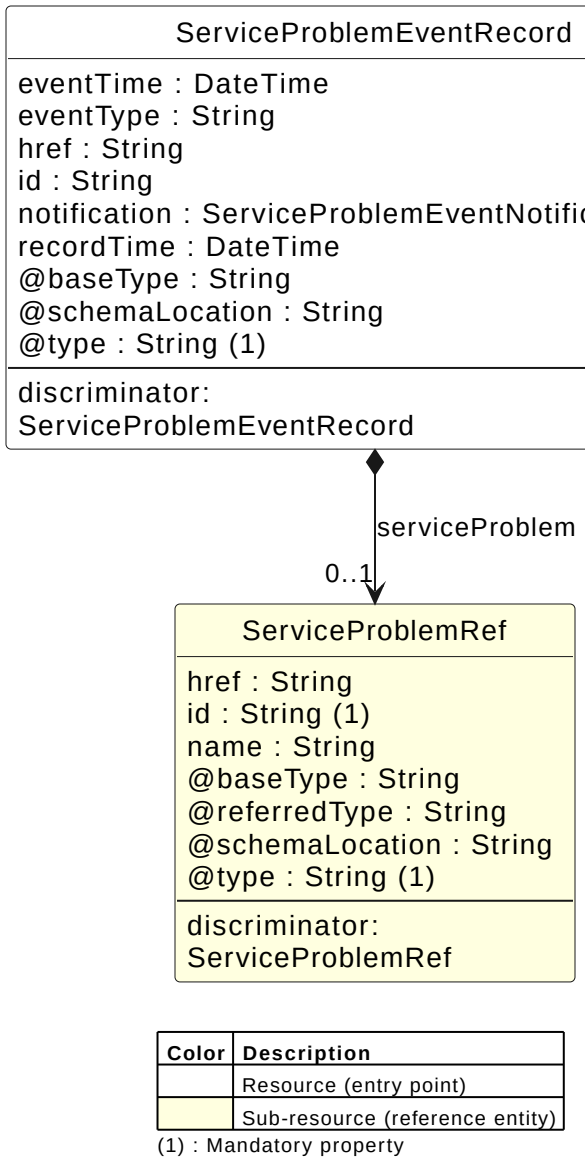


Figure 11 – ServiceProblemEventRecord

Field descriptions

ServiceProblemEventRecord fields

eventTime	A DateTime. Time at which the event occurred.
eventType	A String. Type of the recorded event. This is one of the event types of Service Problem. Example: ServiceProblem CreateEvent, ServiceProblemStateChangeEvent.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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notification	A ServiceProblemEventNotificationType. A notification from the possible Events for Service Problem (such as creation, status change, information required, change).
recordTime	A DateTime. Time at which the record was created.
serviceProblem	A ServiceProblemRef.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemAttributeValueChangeEvent sub-resource fields

ServiceProblemAttributeValueChangeEvent generic structure.

analyticCharacteristic	A Characteristic [0..*]. Collection of analytic characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlationId	A String. The correlation id for this event.
description	A String. An explanation of the event.
domain	A String. The domain of the event.
event	A ServiceProblemAttributeValueChangeEventPayload [1]. ServiceProblemAttributeValueChangeEventPayload generic structure.
eventId	A String [1]. The identifier of the notification.

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eventTime	A DateTime [1]. Time of the event occurrence.
eventType	A String [1]. The type of the notification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
priority	A String. A priority.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties.
reportingSystem	An EntityRef. Entity reference schema to be use for all entityRef class.
source	An EntityRef. Entity reference schema to be use for all entityRef class.
timeOccurred	A DateTime. The time the event occurred.
title	A String. The title of the event.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

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See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

See [BooleanCharacteristic] for details.

FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

See [ObjectArrayCharacteristic] for details.

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valueType sub-resource fields

Characteristic value.

See [valueType] for details.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

See [ObjectCharacteristic] for details.

valueType_2 sub-resource fields

Value of the characteristic.

See [valueType_2] for details.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

See [StringCharacteristic] for details.

ServiceProblemAttributeValueChangeEventPayload sub-resource fields

ServiceProblemAttributeValueChangeEventPayload generic structure.

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serviceProblem	A ServiceProblem_EVO [1]. A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations. ServiceProblem_EVO can be instantiated as * ServiceProblem

ServiceProblem_EVO sub-resource fields

A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations.

affectedLocation	A RelatedPlaceRefOrValue_EVO [0..*]. A list of the locations affected by the problem. At least one of affectedResource, affectedService or affectedLocation should be present.
affectedNumberOfServices	An Integer. Number of affected services.
affectedResource	A ResourceRef_EVO [0..*]. A list of the resources affected by the problem. At least one of affectedResource, affectedService or affectedLocation should be present.
affectedService	A ServiceRef_EVO [0..*]. List of affected services. At least one of affectedResource, affectedService or affected Location should be present.
category	A String. Classifier for the problem. Settable. For example, this is used for distinguish the category of problem originator in [role].[category] format. Example: service Provider.declarer, supplier.originated, system.originated.
characteristic	A Characteristic_EVO [0..*]. A list of additional properties/ data items that describe or characterize this service problem. Please note that this list is currently unconstrained but in a future version this property might be constrained by a specification or deprecated altogether.

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creationDate	A DateTime. Time the problem was created.
description	A String. Free form text describing the Service Problem.
errorMessage	An ErrorMessage_EVO [0..*]. A list of the error(s) associated to the Service Problem. This object is used for any asynchronous errors and error information to be shared with consumer.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. A list of external identifiers assoicated with this problem.
firstAlert	A RelatedEntity_EVO. A reference to an entity, where the type of the entity is not known in advance.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
impactImportanceFactor	An Integer. Impact Importance is characterized by an Impact Importance Factor: overall importance of the impact of all the affected services, e.g. 0 (zero impact) to 100 (worst impact). The Impact Importance is a calculated field which is set by the OSS determining the impact.
impactPattern	An ImpactPattern_EVO. ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.
lastUpdate	A DateTime. Time the problem was last changed.
name	A String. Name of the Service Problem.
note	A Note_EVO [0..*]. A list of comments or notes made on the problem.
originatingSystem	A String. Indicates in which system the problem was generated. Example: CRM Systems, Assurance System, etc.
originatorParty	A RelatedPartyRefOrPartyRoleRef_EVO. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
originatorResource	A ResourceRefOrResourceRoleRef_EVO. Resource reference. A reference to a Resource or to the role that a resource plays in a specific context.

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parentProblem	A ServiceProblemRef_EVO [0..*]. The parent problem to which this problem is attached.
priority	An Integer. An indication varying from 1 (highest) to 10 (lowest) of how important it is for the service provider to correct the Service Problem.
problemEscalation	An Integer. Indicates if this service problem has been escalated or not. Possible values are 0 to 10. A value of zero means no escalation. The meanings of values 1-10 are to be determined by the user of the interface, but they show increasing levels of escalation.
reason	A String. Reason, if known, for the service problem that will assist the Service Provider in understanding the underlying cause or contributing factors, enabling more accurate diagnosis and resolution. This can be set to Unknown, otherwise.
relatedEntity	A RelatedEntity_EVO [0..*]. List of entities associated with this problem. Example: Reference to TroubleTicket, Alarm etc.
relatedEvent	An EventRef_EVO [0..*]. List of events associated to this problem.
relatedParty	A RelatedPartyRefOrPartyRoleRef_EVO [0..*]. List of parties or party roles playing a role within the service problem.
resolutionDate	A DateTime. Time the problem was resolved.
responsibleParty	A RelatedPartyRefOrPartyRoleRef_EVO. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
rootCauseResource	A ResourceRef_EVO [0..*]. Resource(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
rootCauseService	A ServiceRef_EVO [0..*]. Service(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
slaViolation	A SLAViolationRef_EVO [0..*]. A List of SLA violations associated with this problem.

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status	A ServiceProblemStateType. The life cycle status of the service problem, such as acknowledged, rejected, inProgress, resolved etc... ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * resolved * cancelled * closed
statusChangeDate	A DateTime. Time the problem was last status changed.
statusChangeReason	A String. The reason of state change.
trackingRecord	A TrackingRecord_EVO [0..*]. List of tracking records that allow the tracking of modifications on the problem.The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
troubleTicket	A TroubleTicketRef_EVO [0..*]. A list of trouble tickets associated with this problem.
underlyingAlarm	A ResourceAlarmRef_EVO [0..*]. A list of alarms underlying this problem.
underlyingProblem	A ServiceProblemRef_EVO [0..*]. A list of underlying problems. Relevant only if this problem is derived from other problems.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ServiceProblem_EVO can be instantiated as * ServiceProblem
value	This property is present in subclasses

RelatedPlaceRefOrValue_EVO sub-resource fields

Use `role` to distinguish the context of the Place being referenced (such as a `delivery-`

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entrance). In the context of a Ref, use @referredType to distinguish the specific Place sub-type (such as a GeographicAddress). In the context of a Value, use @type to specify the Place sub-type.

place	A PlaceRefOrValue_EVO. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	RelatedPlaceRefOrValue_EVO can be instantiated as * RelatedPlaceRefOrValue
value	This property is present in subclasses

GeographicAddress sub-resource fields

Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanProperty Address.

See [GeographicAddress] for details.

StandardIdentifier sub-resource fields

The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource. It is anticipated that multiple standards can provide definitions for a single entity, e.g., a country identifier can be specified in various standards (e.g., "ISO 3166-1 Alpha 2", "ISO 3166-1 Alpha 3", "ISO 3166-1 Numeric").

See [StandardIdentifier] for details.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce

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platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

See [ExternalIdentifier] for details.

GeographicAddressRelationship sub-resource fields

The GeographicAddressRelationship schema represents a relationship between geographic addresses. It defines the structure for storing information about how two geographic addresses are related to each other within a system.

See [GeographicAddressRelationship] for details.

GeographicLocation sub-resource fields

A GeographicLocation is a pure-virtual super-class to the GeoJSON-aligned geometries of Point (addresses and locations), MultiPoint, LineString (streets, highways and boundaries), MultiLineString and Polygon (countries, provinces, tracts of land). Use the @type attribute to specify which of these is being specified by the geometry attribute.

See [GeographicLocation] for details.

bbox sub-resource fields

A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.

See [bbox] for details.

GeographicLocationRef sub-resource fields

Reference to a Geographic Location.

See [GeographicLocationRef] for details.

GeoJsonPolygon sub-resource fields

A GeoJsonPolygon is an array of linear rings.

See [GeoJsonPolygon] for details.

Polygon sub-resource fields

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GeoJSON: An array of linear rings.

See [Polygon] for details.

polygon sub-resource fields

GeoJSON: An array of linear rings.

See [polygon] for details.

GeoJsonPoint sub-resource fields

A GeoJsonPoint is a single point.

See [GeoJsonPoint] for details.

Point sub-resource fields

GeoJSON: A single position.

See [Point] for details.

position sub-resource fields

GeoJSON: A single position.

See [position] for details.

GeoJsonMultiPolygon sub-resource fields

A GeoJson MultiPolygon is an array of polygons.

See [GeoJsonMultiPolygon] for details.

MultiPolygon sub-resource fields

GeoJSON: A collection of Polygons.

See [MultiPolygon] for details.

multiPolygon sub-resource fields

GeoJSON: An array of polygons.

See [multiPolygon] for details.

GeoJsonMultiPoint sub-resource fields

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GeoJson MultiPoint is a one or more GeoJSON points.

See [GeoJsonMultiPoint] for details.

MultiPoint sub-resource fields

GeoJSON: A collection of Points.

See [MultiPoint] for details.

positionArray sub-resource fields

GeoJSON: An array of positions.

See [positionArray] for details.

GeoJsonMultiLineString sub-resource fields

GeoJson MultiLineString is a one or more GeoJSON LineStrings.

See [GeoJsonMultiLineString] for details.

MultiLineString sub-resource fields

GeoJSON: A collection of distinct LineStrings.

See [MultiLineString] for details.

lineString sub-resource fields

GeoJSON: An array of two or more positions.

See [lineString] for details.

GeoJsonLineString sub-resource fields

GeoJson LineString is a collection of points forming a connected line.

See [GeoJsonLineString] for details.

LineString sub-resource fields

GeoJSON: A collection of Points forming a connected line.

See [LineString] for details.

GeographicSite sub-resource fields

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A GeographicSite is a managed place, such as a retail store, warehouse or data centre. It will have related responsible parties, a calendar of access times and be described by addresses and/or locations.

See [GeographicSite] for details.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

See [CalendarPeriod] for details.

HourPeriod sub-resource fields

Hour interval.

See [HourPeriod] for details.

ContactMedium sub-resource fields

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContact Medium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...

See [ContactMedium] for details.

EmailContactMedium sub-resource fields

Describes an email that could be used to contact a party (an individual or an organization).

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

Describes a fax number that could be used to contact a party (an individual or an organization).

See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

Describes a geographical address that could be used to contact a party (an individual or an

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organization).

See [GeographicAddressContactMedium] for details.

GeographicAddressRef sub-resource fields

Reference to a Geographic Address.

See [GeographicAddressRef] for details.

PhoneContactMedium sub-resource fields

Describes a phone number that could be used to contact a party (an individual or an organization).

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

Describes a social media identifier that could be used to contact a party (an individual or an organization).

See [SocialContactMedium] for details.

WebFormContactMedium sub-resource fields

Describes an URL of a web form that could be used to contact a party (an individual or an organization).

See [WebFormContactMedium] for details.

GeographicSiteRelationship sub-resource fields

Details of a geographic site relationship.

See [GeographicSiteRelationship] for details.

EntityRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [EntityRef] for details.

Note sub-resource fields

Extra information about a given entity.

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See [Note] for details.

RelatedGeographicAddressRef sub-resource fields

A Postal Address (by reference) with its associated role.

See [RelatedGeographicAddressRef] for details.

RelatedGeographicLocationRef sub-resource fields

A Geographic Location (by reference) with its associated role.

See [RelatedGeographicLocationRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

A Party reference.

See [PartyRef] for details.

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

See [PartyRoleRef] for details.

GeographicSiteFeature sub-resource fields

Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.

See [GeographicSiteFeature] for details.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

See [Attachment] for details.

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AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

See [AttachmentRef] for details.

FeatureRelationship sub-resource fields

Configuration feature.

See [FeatureRelationship] for details.

PolicyRef sub-resource fields

Reference to managed Policy object.

See [PolicyRef] for details.

Feature sub-resource fields

Configuration feature.

See [Feature] for details.

GeographicSiteRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [GeographicSiteRef] for details.

GeographicSubAddress sub-resource fields

Representation of a GeographicSubAddress It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.

See [GeographicSubAddress] for details.

GeographicSubAddressUnit sub-resource fields

Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.

See [GeographicSubAddressUnit] for details.

PlaceRef sub-resource fields

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A reference to a physical Place.

See [PlaceRef] for details.

GeoJsonPolygon_EVO sub-resource fields

A GeoJsonPolygon is an array of linear rings.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A Polygon_EVO. GeoJSON: An array of linear rings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier_EVO sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.

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id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ExternalIdentifier_EVO can be instantiated as * ExternalIdentifier
value	This property is present in subclasses

Polygon_EVO sub-resource fields

GeoJSON: An array of linear rings.

coordinates	A polygon_EVO. GeoJSON: An array of linear rings.
type	A String. ENUMERATED with values: * Polygon

GeoJsonPoint_EVO sub-resource fields

A GeoJsonPoint is a single point.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A Point_EVO. GeoJSON: A single position.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Point_EVO sub-resource fields

GeoJSON: A single position.

coordinates	A position_EVO. GeoJSON: A single position.
type	A String. ENUMERATED with values: * Point

GeoJsonMultiPolygon_EVO sub-resource fields

A GeoJson MultiPolygon is an array of polygons.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiPolygon_EVO. GeoJSON: A collection of Polygons.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MultiPolygon_EVO sub-resource fields

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GeoJSON: A collection of Polygons.

coordinates	A multiPolygon_EVO. GeoJSON: An array of polygons.
type	A String. ENUMERATED with values: * MultiPolygon

GeoJsonMultiPoint_EVO sub-resource fields

GeoJson MultiPoint is a one or more GeoJSON points.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiPoint_EVO. GeoJSON: A collection of Points.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MultiPoint_EVO sub-resource fields

GeoJSON: A collection of Points.

coordinates	A positionArray_EVO. GeoJSON: An array of positions.
type	A String. ENUMERATED with values: * MultiPoint

GeoJsonMultiLineString_EVO sub-resource fields

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GeoJson MultiLineString is a one or more GeoJSON LineStrings.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A MultiLineString_EVO. GeoJSON: A collection of distinct LineStrings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

MultiLineString_EVO sub-resource fields

GeoJSON: A collection of distinct LineStrings.

coordinates	A lineString_EVO. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * MultiLineString

GeoJsonLineString_EVO sub-resource fields

GeoJson LineString is a collection of points forming a connected line.

bbox	A bbox_EVO. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.

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description	A String. Additional information of a user-friendly description of the Place.
externalIdentifier	An ExternalIdentifier_EVO [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.
geoJson	A LineString_EVO. GeoJSON: A collection of Points forming a connected line.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A user-friendly name for the Place.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

LineString_EVO sub-resource fields

GeoJSON: A collection of Points forming a connected line.

coordinates	A lineString_EVO. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * LineString

RelatedPlaceRefOrValue sub-resource fields

Use `role` to distinguish the context of the Place being referenced (such as a `delivery-entrance`). In the context of a Ref, use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). In the context of a Value, use `@type` to specify the Place sub-type.

See [RelatedPlaceRefOrValue] for details.

ResourceRef_EVO sub-resource fields

Resource reference, for when Resource is used by other entities.

href	A String. The URI of the referred entity.

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id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ResourceRef_EVO can be instantiated as * ResourceRef
value	This property is present in subclasses

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

See [ResourceRef] for details.

ServiceRef_EVO sub-resource fields

Service reference, for when Service is used by other entities.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ServiceRef_EVO can be instantiated as * ServiceRef
value	This property is present in subclasses

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ServiceRef sub-resource fields

Service reference, for when Service is used by other entities.

See [ServiceRef] for details.

Characteristic_EVO sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

characteristicRelationship	A CharacteristicRelationship_EVO [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Characteristic_EVO can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * Characteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship_EVO sub-resource fields

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Another Characteristic that is related to the current Characteristic;.

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	CharacteristicRelationship_EVO can be instantiated as * CharacteristicRelationship
value	This property is present in subclasses

ErrorMessage_EVO sub-resource fields

Represents an Error.

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ErrorMessage_EVO can be instantiated as * ErrorMessage
value	This property is present in subclasses

ErrorMessage sub-resource fields

Represents an Error.

See [ErrorMessage] for details.

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RelatedEntity_EVO sub-resource fields

A reference to an entity, where the type of the entity is not known in advance.

entity	An EntityRef. Entity reference schema to be use for all entityRef class.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	RelatedEntity_EVO can be instantiated as * RelatedEntity
value	This property is present in subclasses

RelatedEntity sub-resource fields

A reference to an entity, where the type of the entity is not known in advance.

See [RelatedEntity] for details.

ImpactPattern_EVO sub-resource fields

ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.

characteristic	A Characteristic_EVO [0..*]. A generic list of any type of elements. Used for extensions or loose element encapsulation from other namespaces.
description	A String. Basic description of the impact pattern.
id	A String. Locally unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ImpactPattern_EVO can be instantiated as * ImpactPattern
value	This property is present in subclasses

ImpactPattern sub-resource fields

ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.

See [ImpactPattern] for details.

Note_EVO sub-resource fields

Extra information about a given entity.

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	Note_EVO can be instantiated as * Note
value	This property is present in subclasses

RelatedPartyRefOrPartyRoleRef_EVO sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

partyOrPartyRole	A PartyRefOrPartyRoleRef_EVO. A reference to a Party or a PartyRole.

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role	A String. The role the referred Party or PartyRole plays in the specific context of the entity it is linked to, such as 'initiator', 'customer', 'salesAgent', 'user'; NOTE: If used to refer to a PartyRole, the PartyRole describes the role the respective engagedParty plays in a more general context, like e.g. 'Customer', 'Manager'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	RelatedPartyRefOrPartyRoleRef_EVO can be instantiated as * RelatedPartyRefOrPartyRoleRef
value	This property is present in subclasses

ResourceRoleRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [ResourceRoleRef] for details.

ServiceProblemRef_EVO sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ServiceProblemRef_EVO can be instantiated as * ServiceProblemRef

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value	This property is present in subclasses

ServiceProblemRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [ServiceProblemRef] for details.

EventRef_EVO sub-resource fields

Events linked with service problem.

eventTime	A DateTime. Time the event occurred.
href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	EventRef_EVO can be instantiated as * EventRef
value	This property is present in subclasses

EventRef sub-resource fields

Events linked with service problem.

See [EventRef] for details.

SLAViolationRef_EVO sub-resource fields

Entity reference schema to be use for all entityRef class.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.

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name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	SLAViolationRef_EVO can be instantiated as * SLAViolationRef
value	This property is present in subclasses

SLAViolationRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [SLAViolationRef] for details.

TrackingRecord_EVO sub-resource fields

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

characteristic	A Characteristic_EVO [0..*]. A generic list of any type of elements. Used for vendor Extensions or loose element encapsulation from other namespaces.
description	A String. Describes the action being done, such as: ack, clear.
id	A String. Identifier of the TrackingRecord.
systemId	A String. Describes the system Id from which the action was done.
time	A DateTime. Describes the time at which the action was done.
user	A String. Describes the user doing the action.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	TrackingRecord_EVO can be instantiated as * TrackingRecord
value	This property is present in subclasses

TrackingRecord sub-resource fields

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

See [TrackingRecord] for details.

TroubleTicketRef_EVO sub-resource fields

Reference to an TroubleTicket object.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	TroubleTicketRef_EVO can be instantiated as * TroubleTicketRef
value	This property is present in subclasses

TroubleTicketRef sub-resource fields

Reference to an TroubleTicket object.

See [TroubleTicketRef] for details.

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ResourceAlarmRef_EVO sub-resource fields

A set of alarm ids identifying the alarms that are underlying this problem.

changeRequest	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ResourceAlarmRef_EVO can be instantiated as * ResourceAlarmRef
value	This property is present in subclasses

ResourceAlarmRef sub-resource fields

A set of alarm ids identifying the alarms that are underlying this problem.

See [ResourceAlarmRef] for details.

ServiceProblem sub-resource fields

A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations.

affectedLocation	A RelatedPlaceRefOrValue [0..*]. A list of the locations affected by the problem. At least one of affected Resource, affectedService or affectedLocation should be present.
affectedNumberOfServices	An Integer. Number of affected services.

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affectedResource	A ResourceRef [0..*]. A list of the resources affected by the problem. At least one of affectedResource, affected Service or affectedLocation should be present.
affectedService	A ServiceRef [0..*]. List of affected services. At least one of affectedResource, affectedService or affectedLocation should be present.
category	A String. Classifier for the problem. Settable. For example, this is used for distinguish the category of problem originator in [role].[category] format. Example: service Provider.declarer, supplier.originated, system.originated.
characteristic	A Characteristic [0..*]. A list of additional properties/data items that describe or characterize this service problem. Please note that this list is currently unconstrained but in a future version this property might be constrained by a specification or deprecated altogether. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
creationDate	A DateTime. Time the problem was created.
description	A String [1]. Free form text describing the Service Problem.
errorMessage	An ErrorMessage [0..*]. A list of the error(s) associated to the Service Problem. This object is used for any asynchronous errors and error information to be shared with consumer.
externalIdentifier	An ExternalIdentifier [0..*]. A list of external identifiers assoicated with this problem.
firstAlert	A RelatedEntity. A reference to an entity, where the type of the entity is not known in advance.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
impactImportanceFactor	An Integer. Impact Importance is characterized by an Impact Importance Factor: overall importance of the impact of all the affected services, e.g. 0 (zero impact) to 100 (worst impact). The Impact Importance is a calculated field which is set by the OSS determining the impact.
impactPattern	An ImpactPattern. ImpactPattern is a used to capture the nature and scope of the impact caused by a Service Problem. It is used to describe how a service problem affects other services or components, especially when the impact cannot be fully captured by predefined attributes.
lastUpdate	A DateTime. Time the problem was last changed.
name	A String. Name of the Service Problem.
note	A Note [0..*]. A list of comments or notes made on the problem.
originatingSystem	A String. Indicates in which system the problem was generated. Example: CRM Systems, Assurance System, etc.
originatorParty	A RelatedPartyRefOrPartyRoleRef. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
originatorResource	A ResourceRefOrResourceRoleRef. Resource reference. A reference to a Resource or to the role that a resource plays in a specific context.
parentProblem	A ServiceProblemRef [0..*]. The parent problem to which this problem is attached.
priority	An Integer [1]. An indication varying from 1 (highest) to 10 (lowest) of how important it is for the service provider to correct the Service Problem.
problemEscalation	An Integer. Indicates if this service problem has been escalated or not. Possible values are 0 to 10. A value of zero means no escalation. The meanings of values 1-10 are to be determined by the user of the interface, but they show increasing levels of escalation.

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reason	A String [1]. Reason, if known, for the service problem that will assist the Service Provider in understanding the underlying cause or contributing factors, enabling more accurate diagnosis and resolution. This can be set to Unknown, otherwise.
relatedEntity	A RelatedEntity [0..*]. List of entities associated with this problem. Example: Reference to TroubleTicket, Alarm etc.
relatedEvent	An EventRef [0..*]. List of events associated to this problem.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. List of parties or party roles playing a role within the service problem.
resolutionDate	A DateTime. Time the problem was resolved.
responsibleParty	A RelatedPartyRefOrPartyRoleRef. A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.
rootCauseResource	A ResourceRef [0..*]. Resource(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
rootCauseService	A ServiceRef [0..*]. Service(s) that are associated to the underlying service problems that are the Root Cause of this one if any (used only if applicable).
slaViolation	A SLAViolationRef [0..*]. A List of SLA violations associated with this problem.
status	A ServiceProblemStateType. The life cycle status of the service problem, such as acknowledged, rejected, inProgress, resolved etc... ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * resolved * cancelled * closed
statusChangeDate	A DateTime. Time the problem was last status changed.
statusChangeReason	A String. The reason of state change.

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trackingRecord	A TrackingRecord [0..*]. List of tracking records that allow the tracking of modifications on the problem.The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
troubleTicket	A TroubleTicketRef [0..*]. A list of trouble tickets associated with this problem.
underlyingAlarm	A ResourceAlarmRef [0..*]. A list of alarms underlying this problem.
underlyingProblem	A ServiceProblemRef [0..*]. A list of underlying problems. Relevant only if this problem is derived from other problems.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemCreateEvent sub-resource fields

ServiceProblemCreateEvent generic structure.

analyticCharacteristic	A Characteristic [0..*]. Collection of analytic characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlationId	A String. The correlation id for this event.
description	A String. An explanation of the event.

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domain	A String. The domain of the event.
event	A ServiceProblemCreateEventPayload [1]. ServiceProblemCreateEventPayload generic structure.
eventId	A String [1]. The identifier of the notification.
eventTime	A DateTime [1]. Time of the event occurrence.
eventType	A String [1]. The type of the notification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
priority	A String. A priority.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties.
reportingSystem	An EntityRef. Entity reference schema to be use for all entityRef class.
source	An EntityRef. Entity reference schema to be use for all entityRef class.
timeOccurred	A DateTime. The time the event occurred.
title	A String. The title of the event.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemCreateEventPayload sub-resource fields

ServiceProblemCreateEventPayload generic structure.

serviceProblem	A ServiceProblem [1]. A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations.

ServiceProblemDeleteEvent sub-resource fields

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ServiceProblemDeleteEvent generic structure.

analyticCharacteristic	A Characteristic [0..*]. Collection of analytic characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlationId	A String. The correlation id for this event.
description	A String. An explanation of the event.
domain	A String. The domain of the event.
event	A ServiceProblemDeleteEventPayload [1]. ServiceProblemDeleteEventPayload generic structure.
eventId	A String [1]. The identifier of the notification.
eventTime	A DateTime [1]. Time of the event occurrence.
eventType	A String [1]. The type of the notification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
priority	A String. A priority.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties.
reportingSystem	An EntityRef. Entity reference schema to be use for all entityRef class.
source	An EntityRef. Entity reference schema to be use for all entityRef class.
timeOccurred	A DateTime. The time the event occurred.
title	A String. The title of the event.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemDeleteEventPayload sub-resource fields

ServiceProblemDeleteEventPayload generic structure.

serviceProblem	A ServiceProblem_EVO [1]. A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations. ServiceProblem_EVO can be instantiated as * ServiceProblem

ServiceProblemInformationRequiredEvent sub-resource fields

ServiceProblemInformationRequiredEvent generic structure.

analyticCharacteristic	A Characteristic [0..*]. Collection of analytic characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlationId	A String. The correlation id for this event.

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description	A String. An explanation of the event.
domain	A String. The domain of the event.
event	A ServiceProblemInformationRequiredEventPayload [1]. ServiceProblemInformationRequiredEventPayload generic structure.
eventId	A String [1]. The identifier of the notification.
eventTime	A DateTime [1]. Time of the event occurrence.
eventType	A String [1]. The type of the notification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
priority	A String. A priority.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties.
reportingSystem	An EntityRef. Entity reference schema to be use for all entityRef class.
source	An EntityRef. Entity reference schema to be use for all entityRef class.
timeOccurred	A DateTime. The time the event occurred.
title	A String. The title of the event.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemInformationRequiredEventPayload sub-resource fields

ServiceProblemInformationRequiredEventPayload generic structure.

informationRequired	An InformationRequiredArray [1]. InformationRequiredArray generic structure.
serviceProblem	A ServiceProblemRef [1].

ServiceProblemStateChangeEvent sub-resource fields

ServiceProblemStateChangeEvent generic structure.

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analyticCharacteristic	A Characteristic [0..*]. Collection of analytic characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
correlationId	A String. The correlation id for this event.
description	A String. An explanation of the event.
domain	A String. The domain of the event.
event	A ServiceProblemStateChangeEventPayload [1]. ServiceProblemStateChangeEventPayload generic structure.
eventId	A String [1]. The identifier of the notification.
eventTime	A DateTime [1]. Time of the event occurrence.
eventType	A String [1]. The type of the notification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
priority	A String. A priority.
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related parties.
reportingSystem	An EntityRef. Entity reference schema to be use for all entityRef class.
source	An EntityRef. Entity reference schema to be use for all entityRef class.
timeOccurred	A DateTime. The time the event occurred.
title	A String. The title of the event.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemStateChangeEventPayload sub-resource fields

ServiceProblemStateChangeEventPayload generic structure.

serviceProblem	A ServiceProblem_EVO [1]. A Service Problem is an indication that a service is no longer functioning according to the agreement with its client(s). It is a condition where a service fails to meet its expected performance, availability, or quality standards, resulting in a negative impact on the customer experience or business operations. ServiceProblem_EVO can be instantiated as * ServiceProblem

Json representation sample(s)

We provide below a JSON representation as example of the 'ServiceProblemEventRecord' resource object.


```
{
  "id": "42",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblemEventRecord/42",
  "recordTime": "2024-08-08T10:45:30.0Z",
  "eventType": "ServiceProblemCreateEvent",
  "eventTime": "2024-08-08T10:45:25.0Z",
  "@type": "ServiceProblemEventRecord",
  "notification": {
    "eventType": "ServiceProblemCreateEvent",
    "eventTime": "2024-08-08T10:45:25.0Z",
    "eventId": "92775",
    "@type": "ServiceProblemCreateEvent",
    "event": {
      "serviceProblem": {
        "id": "sp_001",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
        "category": "serviceProvider.declared",
        "priority": 1,
        "description": "Internet connection error",
```


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```
        "statusChangeDate": "2024-08-08T10:45:20.0Z",
        "reason": "unknown",
        "status": "acknowledged",
        "originatorParty": {
          "role": "ServiceProvider",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "SP_00001",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/party/SP_00001",
            "name": "ABCompany",
            "@type": "PartyRef",
            "@referredType": "Organization"
          }
        },
        "affectedService": [
          {
            "id": "SP00001_Service_001",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
            "@type": "ServiceRef",
            "@referredType": "Service"
          }
        ],
        "@type": "ServiceProblem"
      }
    }
  },
  "serviceProblem": {
    "id": "sp_001",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  }
}
```


ProblemGroup resource

Task resource that requests Service Problems to be grouped together into a parent and set of children. The problemGroup task is used to group related service problems under a common parent problem so as to enable logical grouping of related issues that may share a common root cause or require coordinated resolution. This improves operational efficiency by consolidating related problems, enhances visibility across incidents, simplifies tracking and reporting, and supports clearer communication among stakeholders. Depending on the implementor's design, the problemGroup task may support partial updates, allowing selective removal of some child problems while retaining others.

Resource model

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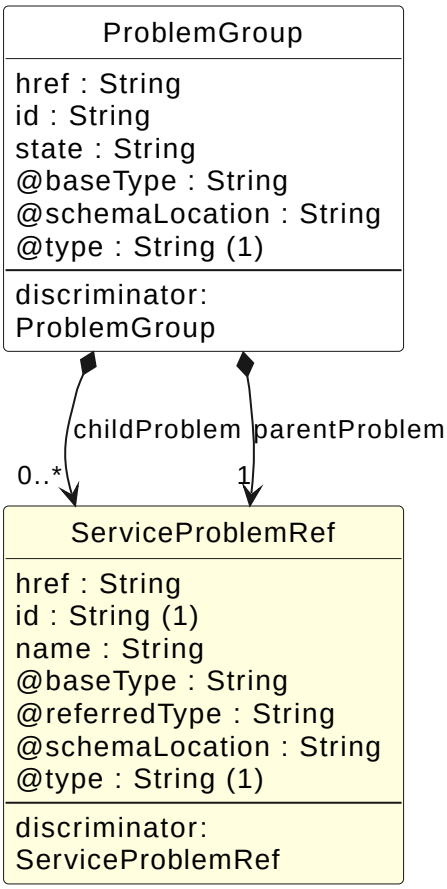


Figure 12 - ProblemGroup

Field descriptions

ProblemGroup fields

childProblem	A ServiceProblemRef [1..*]. ChildProblem is used to associate multiple service problems under a single parent problem. Each childProblem reference can be associated only once with a parentProblem. Depending on the implementor's requirements, a childProblem may either be restricted to a single parent or allowed to be reassigned to a different/multiple parents. Please note both the childProblem and the parentProblem must exist in the system prior to establishing the association.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
parentProblem	A ServiceProblemRef [1].
state	A String. State of the Problem Group task (acknowledged, rejected, inProgress, cancelled, done, terminatedWith Error).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [ServiceProblemRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProblemGroup' resource object.

```
{
  "id": "pg_00001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
```

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```
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v4v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemGroup"
}
```


ProblemUngroup resource

Task resource that requests Service Problems to be ungrouped from a parent. The problem Ungroup task is used to disassociate child service problems from a parent problem within a problemGroup. This operation breaks the logical grouping, allowing individual service problems to be managed independently. For example, it can be applied in the scenarios where the relationship between problems is no longer valid, OR the root cause has been resolved and individual issues need separate handling etc. Depending on the implementor's design, partial ungrouping could be supported. The task can target specific child problems for removal while leaving others intact to allow flexible and selective updates to the problem group structure.

Resource model

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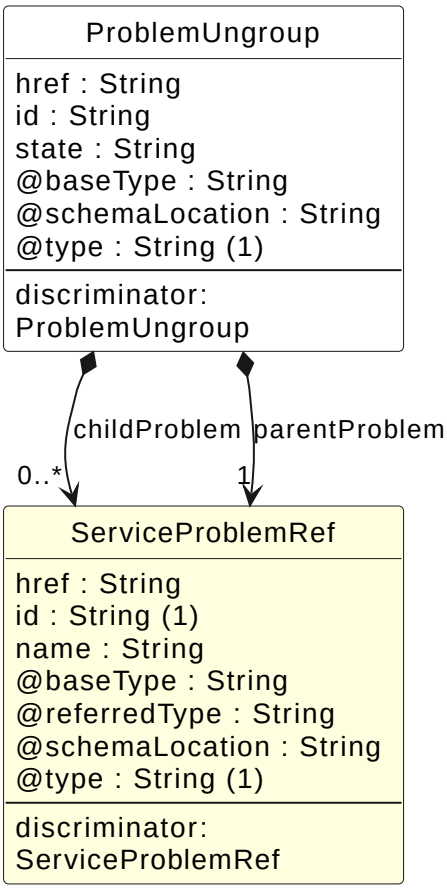


Figure 13 - ProblemUngroup

Field descriptions

ProblemUngroup fields

childProblem	A ServiceProblemRef [1..*]. List of problems to be ungrouped from a parent problem, i.e., the specific service problems to be disassociated from a parent problem within a problemGroup. Only child problems that are currently linked to the specified parent can be ungrouped. Please note both the childProblem and the parentProblem must exist in the system for the ungrouping operation to proceed.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
parentProblem	A ServiceProblemRef [1].
state	A String. State of the Problem Ungroup task (acknowledged, rejected, inProgress, cancelled, done, terminatedWithError).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ServiceProblemRef sub-resource fields

Entity reference schema to be use for all entityRef class.

See [ServiceProblemRef] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProblemUngroup' resource object.


```
{
  "id": "pu_00001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
```


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```
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemUngroup"
}
```


Notification Resource Models

11 notifications are defined for this API.

Notifications related to ServiceProblem:

- Create Event
- Delete Event
- Attribute Value Change Event
- Information Required Event
- State Change Event

Notifications related to ProblemGroup:

- Create Event
- Delete Event
- State Change Event

Notifications related to ProblemUngroup:

- Create Event
- Delete Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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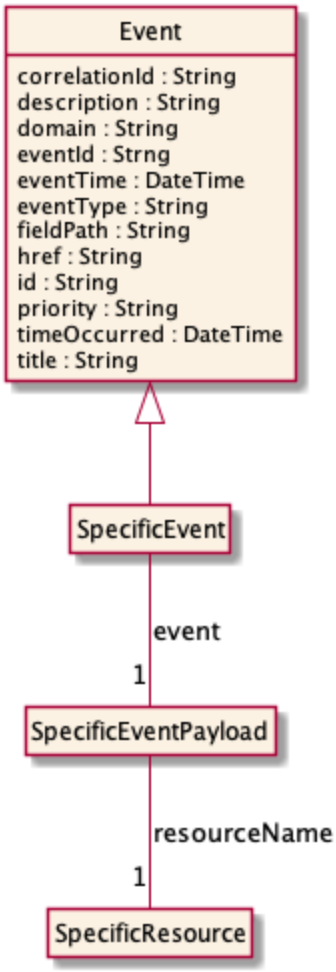


Figure 14 Notification Pattern

ServiceProblem

Create Event

Message example for ServiceProblemCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "2cdbfa6e-6756",
  "description": "ServiceProblemCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "41af-bb8d-b056f61b8696",
  "eventTime": "2024-08-25T12:22:14.903Z",
  "eventType": "ServiceProblemCreateEvent",
  "priority": "5",
  "timeOccurred": "2024-08-25T12:22:12.953Z",
  "title": "ServiceProblemCreateEvent",
  "event": {
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
```

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```
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "category": "serviceProvider.declared",
      "priority": 1,
      "description": "Internet connection error",
      "statusChangeDate": "2024-08-25T12:20:12.953Z",
      "reason": "unknown",
      "status": "acknowledged",
      "originatorParty": {
        "role": "ServiceProvider",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "SP_00001",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/party/SP_00001",
          "name": "ABCompany",
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      },
      "affectedService": [
        {
          "id": "SP00001_Service_001",
          "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
          "@type": "ServiceRef",
          "@referredType": "Service"
        }
      ],
      "@type": "ServiceProblem"
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ServiceProblemCreateEvent"
}
```

Delete Event

Message example for ServiceProblemDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "36af9e61-4a51",
  "description": "ServiceProblemDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4154-a76e-bf08c4bccb2e",
```

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```
  "eventTime": "2022-08-25T12:19:28.549Z",
  "eventType": "ServiceProblemDeleteEvent",
  "priority": "1",
  "timeOccurred": "2022-08-25T12:19:24.023Z",
  "title": "ServiceProblemDeleteEvent",
  "event": {
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "@type": "ServiceProblem"
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ServiceProblemDeleteEvent"
}
```


Attribute Value Change Event

Message example for ServiceProblemAttributeValueChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "b1995ca7-9d12",
  "description": "ServiceProblemAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4f3f-849c-c2ad0d10fbe1",
  "eventTime": "2022-08-25T12:22:14.924Z",
  "eventType": "ServiceProblemAttributeValueChangeEvent",
  "priority": "2",
  "timeOccurred": "2022-08-25T12:22:05.372Z",
  "title": "ServiceProblemAttributeValueChangeEvent",
  "event": {
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "originatorParty": {
        "role": "ServiceProvider",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "SP_00001",
          "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/party/SP_00001",
          "name": "ABCompany",
```


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```
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      },
      "@type": "ServiceProblem"
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ServiceProblemAttributeValueChangeEvent"
}
```


Information Required Event

Message example for ServiceProblemInformationRequiredEvent event


```
Content-Type: application/json
{
  "correlationId": "67b04b18-acdf",
  "description": "ServiceProblemInformationRequiredEvent illustration",
  "domain": "Commercial",
  "eventId": "4584-8713-56b25db8715f",
  "eventTime": "2022-08-25T12:22:14.935Z",
  "eventType": "ServiceProblemInformationRequiredEvent",
  "priority": "3",
  "timeOccurred": "2022-08-25T12:22:09.809Z",
  "title": "ServiceProblemInformationRequiredEvent",
  "event": {
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "informationRequired": [
      {
        "op": "add",
        "path": "priority"
      }
    ]
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
```


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```
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ServiceProblemInformationRequiredEvent"
}
```


State Change Event

Message example for ServiceProblemStateChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "db69b15f-0b23",
  "description": "ServiceProblemStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4e07-aca3-682fd107b18f",
  "eventTime": "2022-08-25T12:22:14.913Z",
  "eventType": "ServiceProblemStateChangeEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:22:07.094Z",
  "title": "ServiceProblemStateChangeEvent",
  "event": {
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "status": "resolved",
      "@type": "ServiceProblem"
    }
  },
  "reportingSystem": {
    "id": "819",
    "name": "APP-913",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "152",
    "name": "APP-335",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ServiceProblemStateChangeEvent"
}
```


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ProblemGroup

Create Event

Message example for ProblemGroupCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemGroupCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemGroupCreateEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemGroupCreateEvent",
  "event": {
    "problemGroup": {
      "id": "pg_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
      "state": "acknowledged",
      "childProblem": [
        {
          "id": "41",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        },
        {
          "id": "42",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        },
        {
          "id": "43",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        }
      ],
      "parentProblem": {
        "id": "40",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      "@type": "ProblemGroup"
    }
  },
  "reportingSystem": {
```

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```
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemGroupCreateEvent"
}
```

Delete Event

Message example for ProblemGroupDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemGroupDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemGroupDeleteEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemGroupDeleteEvent",
  "event": {
    "problemGroup": {
      "id": "pg_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
      "@type": "ProblemGroup"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemGroupDeleteEvent"
}
```

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State Change Event

Message example for ProblemGroupStateChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemGroupStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemGroupStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemGroupStateChangeEvent",
  "event": {
    "problemGroup": {
      "id": "pg_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
      "@type": "ProblemGroup",
      "state": "done"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemGroupStateChangeEvent"
}
```

ProblemUngroup

Create Event

Message example for ProblemUngroupCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemUngroupCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemUngroupCreateEvent",
```

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```
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemUngroupCreateEvent",
  "event": {
    "problemUngroup": {
      "id": "pu_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
      "state": "acknowledged",
      "childProblem": [
        {
          "id": "41",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        },
        {
          "id": "42",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        },
        {
          "id": "43",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
          "@type": "ServiceProblemRef",
          "@referredType": "ServiceProblem"
        }
      ],
      "parentProblem": {
        "id": "40",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      "@type": "ProblemUngroup"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemUngroupCreateEvent"
}
```


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Delete Event

Message example for ProblemUngroupDeleteEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemUngroupDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemUngroupDeleteEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemUngroupDeleteEvent",
  "event": {
    "problemUngroup": {
      "id": "pu_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
      "@type": "ProblemUngroup"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemUngroupDeleteEvent"
}
```

State Change Event

Message example for ProblemUngroupStateChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "b122fa39-7242",
  "description": "ProblemUngroupStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "4d6e-98cf-a085d0121f61",
  "eventTime": "2022-08-25T12:20:45.491Z",
  "eventType": "ProblemUngroupStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:20:44.232Z",
  "title": "ProblemUngroupStateChangeEvent",
  "event": {
```

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```
    "problemUngroup": {
      "id": "pu_00001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
      "state": "done",
      "@type": "ProblemUngroup"
    }
  },
  "reportingSystem": {
    "id": "768",
    "name": "APP-897",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "276",
    "name": "APP-819",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "ProblemUngroupStateChangeEvent"
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ServiceProblem

Retrieves a serviceProblem by ID

`GET /serviceProblem/{id}?fields=…`

Description

This operation retrieves a ServiceProblem entity. Attribute selection is enabled for all first level attributes.

Usage samples

Here's an example of a request for retrieving a specific service problem.

Request

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```
GET /serviceProblem/problemxxxx0000
Content-Type: application/json
```

Response

```
200

{
  "id": "problemxxxx0000",
  "originatingSystem": "System_001",
  "category": "supplier.originated",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0000",
  "impactImportanceFactor": 0,
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka",
  "problemEscalation": 0,
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "statusChangeReason": "problem analysis has been completed in NP1",
  "resolutionDate": "2017-10-29T12:00:00.361Z",
  "status": "resolved",
  "reason": "Failure of resource NP1_Resource_1 in NP1",
  "affectedNumberOfServices": 2,
  "firstAlert": {
    "@type": "RelatedEntity",
    "role": "trigger",
    "entity": {
      "@type": "EntityRef",
      "@referredType": "TroubleTicket",
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/troubleTicketManagement/v5/troubleTicket/NP1_TT_000000"
    }
  },
  "responsibleParty": {
    "role": "Customer",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "@type": "PartyRef",
      "id": "456",
      "href": "http://serverlocation:port/tmf-api/partyManagement/v5/individual/456",
      "name": "Jean Pontus",
      "@referredType": "Individual"
    }
  },
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
```

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```
  },
  "relatedParty": [
    {
      "role": "Supplier",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "789",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/789",
        "name": "NP1",
        "@referredType": "Organization"
      }
    },
    {
      "role": "Partner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "123",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/123",
        "name": "SP1",
        "@referredType": "Organization"
      }
    },
    {
      "role": "Partner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "321",
        "href": "http://serverlocation:port/tmf-
api/partyManagement/v5/organization/321",
        "name": "SP2",
        "@referredType": "Organization"
      }
    }
  ],
  "affectedService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
      "@type": "ServiceRef",
      "@referredType": "Service"
    },
    {
      "id": "NP1_Tokyo_xxxx",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_xxxx",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "affectedResource": [
    {
      "id": "NP1_RES_0001",
      "href": "https://mycsp.com:8080/tmf-
api/resourceInventory/v5/resource/NP1_RES_0001",
```


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```
      "@type": "ResourceRef",
      "@referredType": "VirtualResource"
    }
  ],
  "affectedLocation": [
    {
      "role": "VPN Endpoint",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "id": "Loc000000",
        "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/Loc000000/",
        "name": "144 Main Street Tokyo 51663556"
      }
    },
    {
      "role": "VPN Endpoint",
      "@type": "RelatedPlaceRefOrValue",
      "place": {
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "id": "Loc000001",
        "href": "https://mycsp.com:8080/tmf-
api/geographicAddressManagement/v5/geographicAddress/Loc000001/",
        "name": "351 Main Street Osaka 3546365"
      }
    }
  ],
  "troubleTicket": [
    {
      "id": "NP1_TT_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/troubleTicketManagement/v5/TroubleTicketRef/NP1_TT_000000",
      "@type": "TroubleTicketRef",
      "@referredType": "TroubleTicket"
    }
  ],
  "underlyingAlarm": [
    {
      "id": "NP1_A_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/alarmManagement/v5/resourceAlarm/NP1_A_000000",
      "@referredType": "ResourceAlarm",
      "@type": "ResourceAlarmRef"
    }
  ],
  "slaViolation": [
    {
      "id": "NP1_SLA_0000000",
      "href": "https://mycsp.com:8080/tmf-
api/SLAManagement/v5/SLAViolationRef/NP1_SLA_000000",
      "@type": "SLAViolationRef",
      "@referredType": "SLAViolation"
    }
  ],
  "relatedEvent": [
    {
      "id": "prediction_0001",
```


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```
      "href": "https://mycsp.com:8080/tmf-
api/eventManagement/v5/event/prediction_0001",
      "eventTime": "2014-12-20T17:00:00Z",
      "@type": "EventRef",
      "@referredType": "Event"
    }
  ],
  "rootCauseService": [
    {
      "id": "NP1_Tokyo_Osaka",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/NW1_Tokyo_Osaka",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "rootCauseResource": [
    {
      "id": "NP1_Resource_1",
      "href": "https://mycsp.com:8080/tmf-
api/resourceInventory/v5/resource/NP1_Resource_1",
      "@type": "ResourceRef",
      "@referredType": "VNF"
    }
  ],
  "parentProblem": [
    {
      "id": "problemxxxx0001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "underlyingProblem": [
    {
      "id": "problemxxxx0001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/problemxxxx0001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "trackingRecord": [
    {
      "description": "System cleared the problem",
      "systemId": "SNOW",
      "time": "2014-12-20T17:00:00Z",
      "user": "NPUSER1",
      "id": "TR_0001",
      "@type": "TrackingRecord"
    }
  ],
  "note": [
    {
      "id": "2241",
      "author": "SPM IT Team",
      "date": "2014-12-20T17:00:00Z",
      "text": "receive trouble ticket from NP1, and create this Service Problem",
      "@type": "Note"
```


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```
    }
  ],
  "impactPattern": {
    "description": "Many services are at risk in this problem",
    "id": "TR_0002",
    "characteristic": [
      {
        "id": "CH_003",
        "name": "ServiceDisruption",
        "value": true,
        "valueType": "boolean",
        "@type": "BooleanCharacteristic"
      }
    ],
    "@type": "ImpactPattern"
  },
  "@type": "ServiceProblem"
}
```


List or find serviceProblem objects

`GET /serviceProblem?fields=…&{filtering}`

Description

This operation list ServiceProblem entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple service problems, in this example closed items and priority 1 are filtered

Request

```
GET /serviceProblem?filter=status=closed&priority=1
Content-Type: application/json
```

Response


```
200

[
  {
    "id": "sp_001",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
    "category": "serviceProvider.declared",
    "priority": 1,
```


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```
    "description": "Internet connection error",
    "statusChangeDate": "2017-10-29T12:00:00.361Z",
    "reason": "unknown",
    "status": "closed",
    "originatorParty": {
      "role": "ServiceProvider",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "SP_00001",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
        "name": "ABCompany",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    },
    "affectedService": [
      {
        "id": "SP00001_Service_001",
        "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
        "@type": "ServiceRef",
        "@referredType": "Service"
      }
    ],
    "@type": "ServiceProblem"
  },
  {
    "id": "sp_002",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_002",
    "category": "serviceProvider.declared",
    "priority": 1,
    "description": "Wireless connection error",
    "statusChangeDate": "2017-10-29T12:00:00.361Z",
    "reason": "unknown",
    "status": "closed",
    "originatorParty": {
      "role": "ServiceProvider",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "SP_00001",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
        "name": "ABCompany",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    },
    "affectedService": [
      {
        "id": "SP00001_Service_002",
        "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_002",
        "@type": "ServiceRef",
        "@referredType": "Service"
      }
    ],
    "@type": "ServiceProblem"
  }
```


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```
]
```

Creates a serviceProblem

`POST /serviceProblem?fields=…&{filtering}`

Description

This operation creates a ServiceProblem entity.

Mandatory Attributes

Mandatory Attributes	Rule
description	
priority	
reason	
@type	

Usage samples

Here's an example of a request for creating a service problem

Request

```
POST /serviceProblem
Content-Type: application/json

{
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "Internet connection error",
  "reason": "unknown",
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
```

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```
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "@type": "ServiceProblem"
}
```

Response

```
201

{
  "id": "sp_001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "Internet connection error",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "reason": "unknown",
  "status": "acknowledged",
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "@type": "ServiceProblem"
}
```

Updates partially a serviceProblem

`PATCH /serviceProblem/{id}?fields=…`

Description

This operation allows partial updates of a ServiceProblem entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch

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(http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
errorMessage	
firstAlert	
href	
id	
originatingSystem	
trackingRecord	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
affectedLocation	
affectedNumberOfServices	
affectedResource	
affectedService	
category	
characteristic	
creationDate	
description	
externalIdentifier	
impactImportanceFactor	
impactPattern	
lastUpdate	
name	
note	

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Patchable Attributes	Rule
originatorParty	
originatorResource	
parentProblem	
priority	
problemEscalation	
reason	
relatedEntity	
relatedEvent	
relatedParty	
resolutionDate	
responsibleParty	
rootCauseResource	
rootCauseService	
slaViolation	
status	
statusChangeDate	
statusChangeReason	
troubleTicket	
underlyingAlarm	
underlyingProblem	

Usage samples

Here's an example of a request for patching a service problem. In this example, a new description is set.

Request

```
PATCH /serviceProblem/problemxxxx0000
Content-Type: application/json

{
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "@type": "ServiceProblem"
```

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```
}
```

Response

```
200

{
  "id": "sp_001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "reason": "unknown",
  "status": "acknowledged",
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "@type": "ServiceProblem"
}
```

Here's an example of a request for patching a service problem. In this example, a new description is set.

Request

```
PATCH /serviceProblem/problemxxxx0000
Content-Type: application/merge-patch+json

{
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "@type": "ServiceProblem"
}
```

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Response

```
200

{
  "id": "sp_001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "reason": "unknown",
  "status": "acknowledged",
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "@type": "ServiceProblem"
}
```

Here's an example of a request for patching a service problem. In this example, an affected service is added.

Request

```
PATCH /serviceProblem/problemxxxx0000
Content-Type: application/json-patch+json

[
  {
    "op": "add",
    "path": "/affectedService",
    "value": {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventory/v5/service/44",
      "@type": "ServiceRef",
      "@referredType": "Service"
```

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```
    }
  }
]
```

Response

```
200

{
  "id": "sp_001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "reason": "unknown",
  "status": "acknowledged",
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
      "@type": "ServiceRef",
      "@referredType": "Service"
    },
    {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventory/v5/service/44",
      "@type": "ServiceRef",
      "@referredType": "Service"
    }
  ],
  "@type": "ServiceProblem"
}
```

Here's an example of a request for patching a service problem. In this example, a characterstic is added.

Request

```
PATCH /serviceProblem/problemxxxx0000
```

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```
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/characteristic/value?characteristic.name=isIsolated",
    "value": false
  }
]
```

Response

```
200

{
  "id": "sp_001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
  "category": "serviceProvider.declared",
  "priority": 1,
  "description": "connection failure between Tokyo and Osaka at 5:00",
  "statusChangeDate": "2017-10-29T12:00:00.361Z",
  "reason": "unknown",
  "status": "acknowledged",
  "characteristic": [
    {
      "name": "isIsolated",
      "value": false,
      "valueType": "boolean",
      "@type": "BooleanCharacteristic"
    }
  ],
  "originatorParty": {
    "role": "ServiceProvider",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "SP_00001",
      "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organization/SP_00001",
      "name": "ABCompany",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  },
  "affectedService": [
    {
      "id": "SP00001_Service_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
      "@type": "ServiceRef",
      "@referredType": "Service"
    },
    {
      "id": "44",
      "href": "https://mycsp.com:8080/tmf-api/serviceInventory/v5/service/44",
      "@type": "ServiceRef",
      "@referredType": "Service"
```

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```
    }
  ],
  "@type": "ServiceProblem"
}
```

Deletes a serviceProblem

DELETE /serviceProblem/{id}?{filtering}

Description

This operation deletes a ServiceProblem entity.

Operations on ServiceProblemEventRecord

Retrieves a serviceProblemEventRecord by ID

GET /serviceProblemEventRecord/{id}?fields=…

Description

This operation retrieves a ServiceProblemEventRecord entity. Attribute selection is enabled for all first level attributes.

Usage samples

Here's an example of a request for retrieving a specific service problem event record.

Request

```
GET /serviceProblemEventRecord/42
Content-Type: application/json
```

Response

```
200

{
  "id": "42",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblemEventRecord/42",
  "recordTime": "2016-08-08T10:45:30.0Z",
  "eventType": "ServiceProblemCreateEvent",
  "eventTime": "2016-08-08T10:45:25.0Z",
  "notification": {
    "eventType": "ServiceProblemCreateEvent",
    "eventTime": "2024-10-30T10:45:25.0Z",
    "eventId": "92775",
```

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```
    "event": {
      "serviceProblem": {
        "id": "sp_001",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
        "category": "serviceProvider.declared",
        "priority": 1,
        "description": "Internet connection error",
        "creationDate": "2024-10-28T09:00:00.000Z",
        "lastUpdate": "2024-10-29T12:00:00.361Z",
        "statusChangeDate": "2024-10-29T12:00:00.361Z",
        "reason": "unknown",
        "status": "acknowledged",
        "originatorParty": {
          "role": "ServiceProvider",
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "partyOrPartyRole": {
            "id": "SP_00001",
            "href": "https://mycsp.com:8080/tmf-api/partyManagement/v5/party/SP_00001",
            "name": "ABCompany",
            "@type": "PartyRef",
            "@referredType": "Organization"
          }
        },
        "affectedService": [
          {
            "id": "SP00001_Service_001",
            "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
            "@type": "ServiceRef",
            "@referredType": "Service"
          }
        ],
        "@type": "ServiceProblem"
      }
    },
    "@type": "ServiceProblemCreateEvent"
  },
  "serviceProblem": {
    "id": "sp_001",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ServiceProblemEventRecord"
}
```


List or find serviceProblemEventRecord objects

`GET /serviceProblemEventRecord?fields=…&{filtering}`

Description

This operation list ServiceProblemEventRecord entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level

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supported by an implementation.

Usage samples

Here's an example of a request for searching for any event records related to a particular service problem.

Request

```
GET /serviceProblemEventRecord?filter=serviceProblem.id=sp_001
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "42",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblemEventRecord/42",
    "recordTime": "2016-08-08T10:45:30.0Z",
    "eventType": "ServiceProblemCreateEvent",
    "eventTime": "2016-08-08T10:45:25.0Z",
    "notification": {
      "eventType": "ServiceProblemCreateEvent",
      "eventTime": "2024-10-30T10:45:25.0Z",
      "eventId": "92775",
      "event": {
        "serviceProblem": {
          "id": "sp_001",
          "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
          "category": "serviceProvider.declared",
          "priority": 1,
          "description": "Internet connection error",
          "creationDate": "2024-10-28T09:00:00.000Z",
          "lastUpdate": "2024-10-29T12:00:00.361Z",
          "statusChangeDate": "2024-10-29T12:00:00.361Z",
          "reason": "unknown",
          "status": "acknowledged",
          "originatorParty": {
            "role": "ServiceProvider",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
              "id": "SP_00001",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/party/SP_00001",
              "name": "ABCompany",
              "@type": "PartyRef",
              "@referredType": "Organization"
            }
          },
          "affectedService": [
```

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```
            {
              "id": "SP00001_Service_001",
              "href": "https://mycsp.com:8080/tmf-
api/serviceInventory/v5/service/SP00001_Service_001",
              "@type": "ServiceRef",
              "@referredType": "Service"
            }
          ],
          "@type": "ServiceProblem"
        }
      },
      "@type": "ServiceProblemCreateEvent"
    },
    "serviceProblem": {
      "id": "sp_001",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/sp_001",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "@type": "ServiceProblemEventRecord"
  }
]
```

Operations on ProblemGroup

Retrieves a problemGroup by ID

`GET /problemGroup/{id}?fields=…`

Description

This operation retrieves a ProblemGroup entity. Attribute selection is enabled for all first level attributes.

Usage samples

Here's an example of a request for retrieving a specific problem group task.

Request

```
GET /problemGroup/pg_00001
Content-Type: application/json
```

Response

```
200

{
  "id": "pg_00001",
```

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```
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemGroup"
}
```

List or find problemGroup objects

`GET /problemGroup?fields=…&{filtering}`

Description

This operation list ProblemGroup entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple problem group entities

Request

```
GET /problemGroup
```

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```
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "pg_00001",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
    "state": "acknowledged",
    "childProblem": [
      {
        "id": "41",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "42",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "43",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      }
    ],
    "parentProblem": {
      "id": "40",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "@type": "ProblemGroup"
  },
  {
    "id": "pg_00002",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00002",
    "state": "acknowledged",
    "childProblem": [
      {
        "id": "31",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/31",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
```

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```
      {
        "id": "32",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/32",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "33",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/33",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      }
    ],
    "parentProblem": {
      "id": "30",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/30",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "@type": "ProblemGroup"
  }
]
```


Creates a problemGroup

`POST /problemGroup?fields=…&{filtering}`

Description

This operation creates a ProblemGroup entity.

Mandatory Attributes

Mandatory Attributes	Rule
childProblem	
parentProblem	
@type	

Usage samples

Here's an example of a request for creating a problem group task, where the task is long running and is persisted by the API provider. When the task can be executed immediately by the API provider, then return response code should be 200.

Request

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```
POST /problemGroup
Content-Type: application/json

{
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemGroup"
}
```

Response

```
201

{
  "id": "pg_00001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemGroup/pg_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
```

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```
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemGroup"
}
```


Operations on ProblemUngroup

Retrieves a problemUngroup by ID

`GET /problemUngroup/{id}?fields=…`

Description

This operation retrieves a ProblemUngroup entity. Attribute selection is enabled for all first level attributes.

Usage samples

Here's an example of a request for retrieving a specific problem ungroup task.

Request

```
GET /problemUngroup/pu_00001
Content-Type: application/json
```

Response

```
200

{
  "id": "pu_00001",
  "href": "https://mycsp.com:8080/tmf-
```

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```
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemUngroup"
}
```


List or find problemUngroup objects

`GET /problemUngroup?fields=…&{filtering}`

Description

This operation list ProblemUngroup entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple problem ungroup entities

Request

```
GET /problemUngroup
```

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```
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "pu_00001",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
    "state": "acknowledged",
    "childProblem": [
      {
        "id": "41",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "42",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "43",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      }
    ],
    "parentProblem": {
      "id": "40",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "@type": "ProblemUngroup"
  },
  {
    "id": "pu_00002",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
    "state": "acknowledged",
    "childProblem": [
      {
        "id": "31",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/31",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
```

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```
      {
        "id": "32",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/32",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      },
      {
        "id": "33",
        "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/33",
        "@type": "ServiceProblemRef",
        "@referredType": "ServiceProblem"
      }
    ],
    "parentProblem": {
      "id": "30",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/30",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    "@type": "ProblemUngroup"
  }
]
```


Creates a problemUngroup

`POST /problemUngroup?fields=…&{filtering}`

Description

This operation creates a ProblemUngroup entity.

Mandatory Attributes

Mandatory Attributes	Rule
childProblem	
parentProblem	
@type	

Usage samples

Here's an example of a request for creating a problem ungroup task, where the task is long running and is persisted by the API provider. When the task can be executed immediately by the API provider, then return response code should be 200.

Request

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```
POST /problemUngroup
Content-Type: application/json

{
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemUngroup"
}
```

Response

```
201

{
  "id": "pu_00001",
  "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/problemUngroup/pu_00001",
  "state": "acknowledged",
  "childProblem": [
    {
      "id": "41",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/41",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "42",
```

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```
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/42",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    },
    {
      "id": "43",
      "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/43",
      "@type": "ServiceProblemRef",
      "@referredType": "ServiceProblem"
    }
  ],
  "parentProblem": {
    "id": "40",
    "href": "https://mycsp.com:8080/tmf-
api/serviceProblemManagement/v5/serviceProblem/40",
    "@type": "ServiceProblemRef",
    "@referredType": "ServiceProblem"
  },
  "@type": "ProblemUngroup"
}
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	March 2023	Anu Aulakh Telstra anu.aulakh@team.telstra.com	Upgrade to V5 design patterns and OAS3
Version 5.1.0	September 2025	Sheilja Bajaj Telstra sheilja.bajaj@team.telstra.com	Updates to include concrete classes for characteristic and place, remove ProblemAcknowledgement and ProblemUnacknowledgement resources and schema alignment for ServiceProblemEventRecord

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